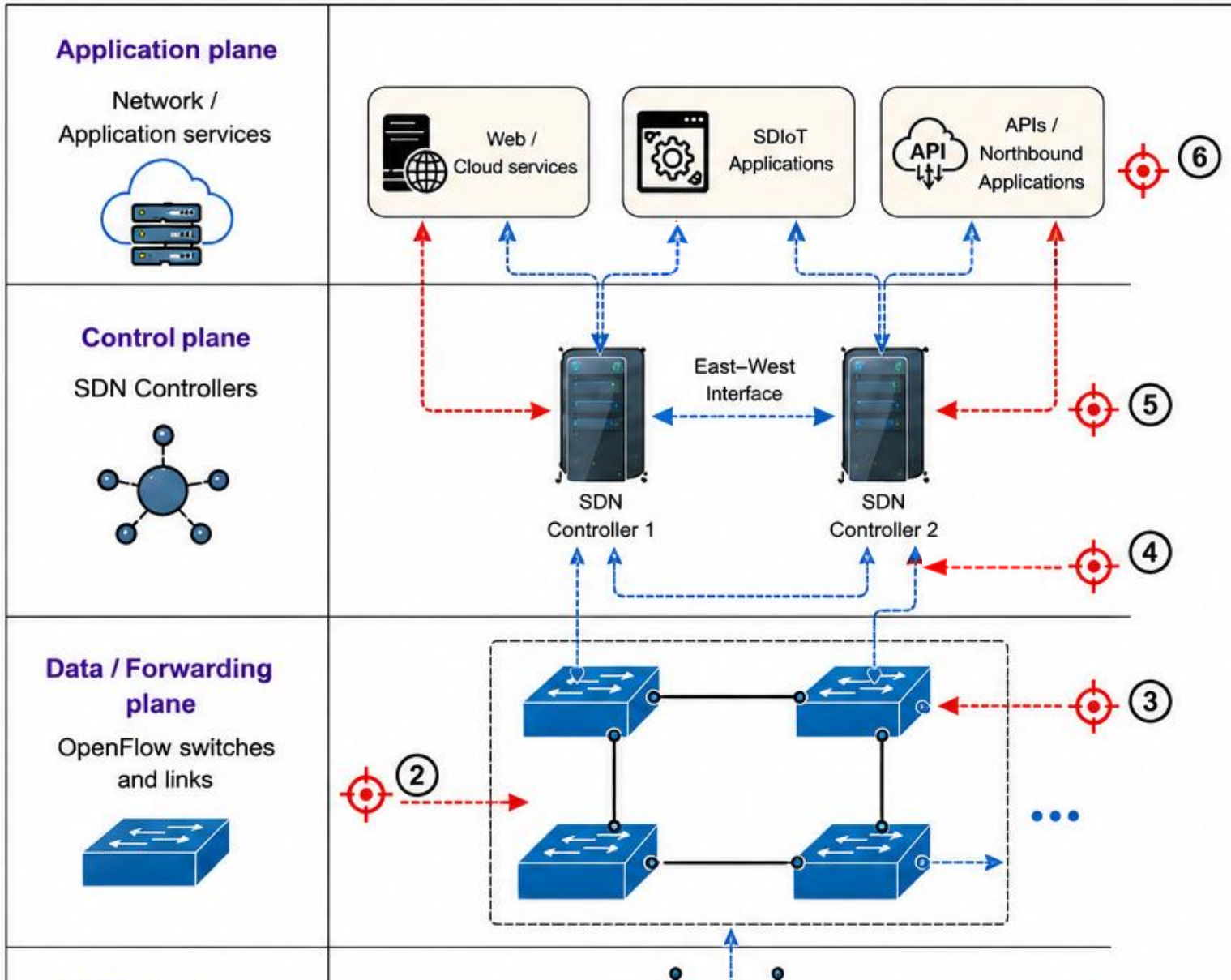
Data Encryption Standard (DES)



ID	Attack Target Point (Where the attack acts)	Representative DDoS Mechanisms (Examples)	Potential In (Exampl
6	Application Layer / Services	• HTTP flooding • Slowloris / Slow HTTP • Application-layer DDoS • API abuse	• Service unava • Increased resp • QoS degradat • Application ou
5	Control Plane / SDN Controller	• Packet-In flooding • Control-plane flooding • Controller overload • Flow-Mod flooding	• Controller CPU exhaustion • Delayed respo • Delayed flow ir • Network instab
4	Southbound Interface (Switch–Controller Channel)	• Excessive Packet-In requests • Flow setup flooding • Control message flood	• Increased cont • Link congestio • Control-plane • Throughput re
3	Data/Forwarding Plane (Switch Resources)	• Flow-table saturation • Flow setup flooding • UDP flooding • ICMP flooding	• Flow-table exh • Increased late • Packet loss • Forwarding de
2	Network Links	• Link flooding • Volumetric flooding • UDP flooding • ICMP flooding	• Bandwidth ex • Link congestio • Packet loss • Path instability

A(n) _____ is a keyless substitution cipher with N inputs and M outputs that uses a formula to define the relationship between the input stream and the output stream.

A) S-box
B) P-box
C) T-box
D) none of the above

19 A(n) _____is a keyless transposition cipher with N inputs and M outputs that uses a table to define the relationship between the input stream and the output stream.

A) S-box
B) P-box
C) T-box
D) none of the above

20 A modern cipher is usually a complex _____cipher made of a combination of different simple ciphers.

A) round
B) circle
C) square
D) none of the above

List the three Characteristics of Cryptosystem

Characteristic of Cryptographic Systems

1. **The type of operations used for transforming plaintext to cipher text.**

   substitution, transposition, product systems

2. **The number of keys used.**

   single-key, secret-key, or conventional encryption

   two-key, or public-key encryption

3. **The way in which the plaintext is processed.**

   block cipher processes, stream cipher processes

Type of operations used for transforming plaintext to cipher text

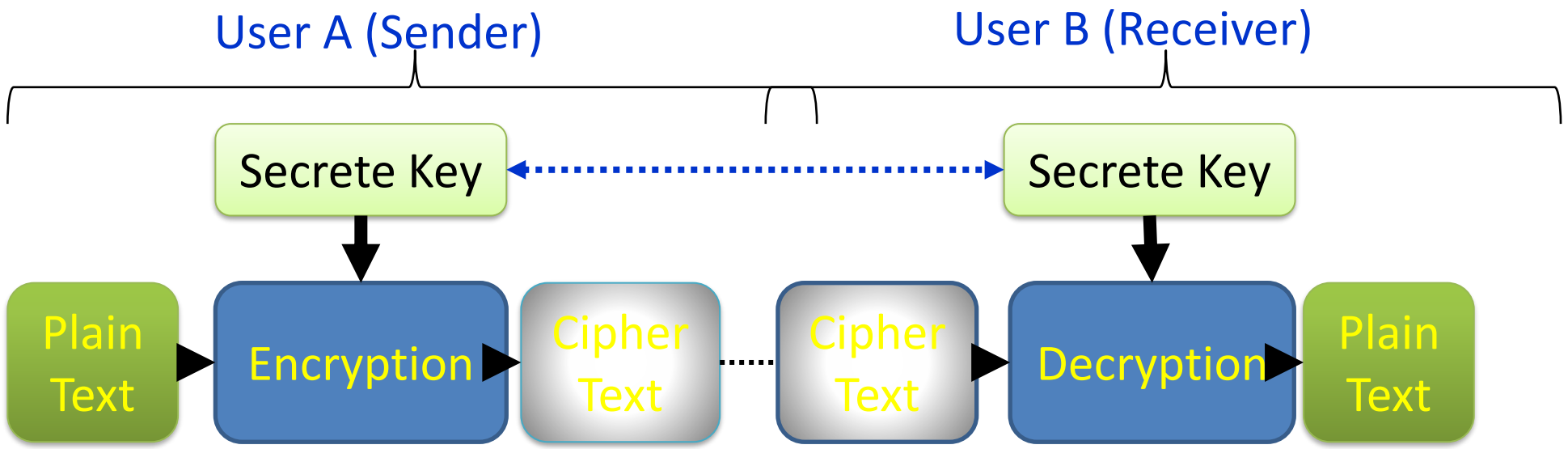
Substitution

In plain text, each character is replaced with another character

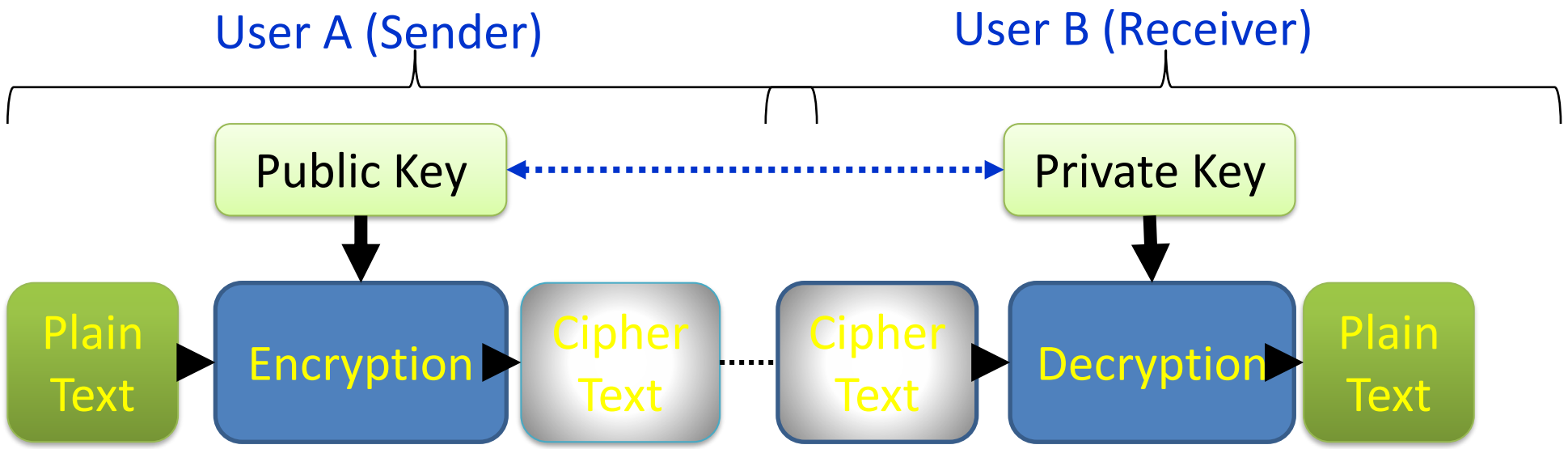
Transposition

Some sort of permutation on the plain text

Symmetric (Private) Key Encryption

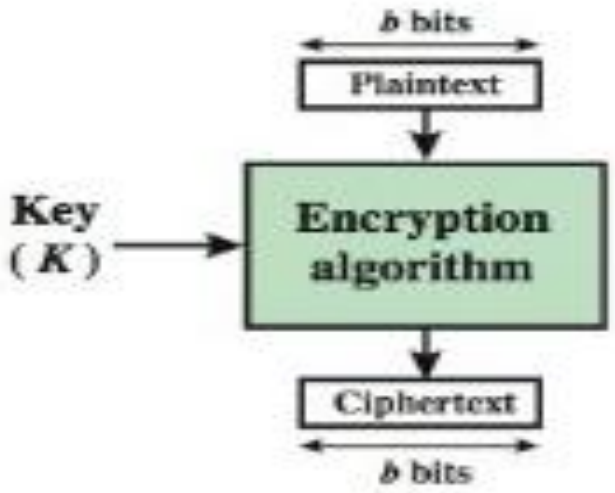


Public Private Key Encryption

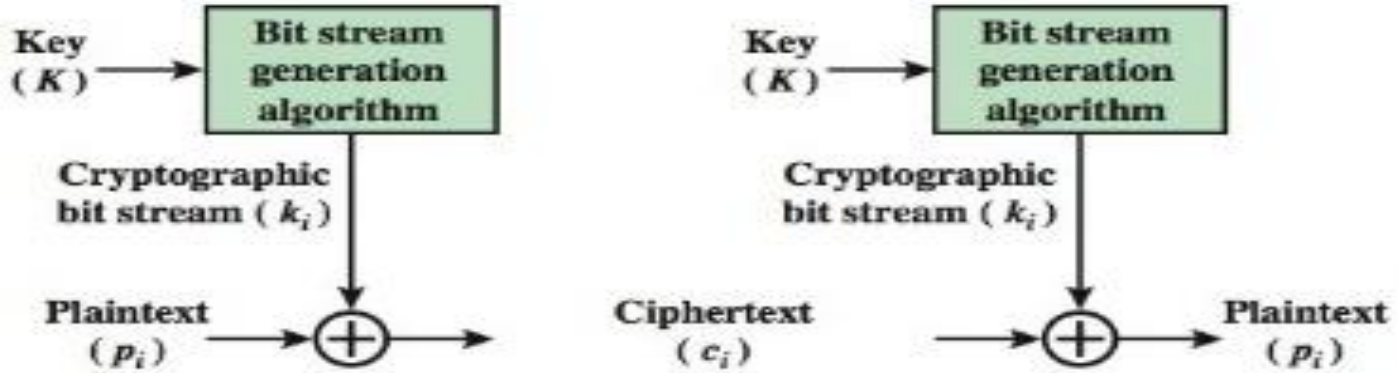


Block Cipher Vs Stream Cipher Principles

process messages in blocks, each of which is then en/decrypted
process messages a bit or byte at a time when en/decrypting



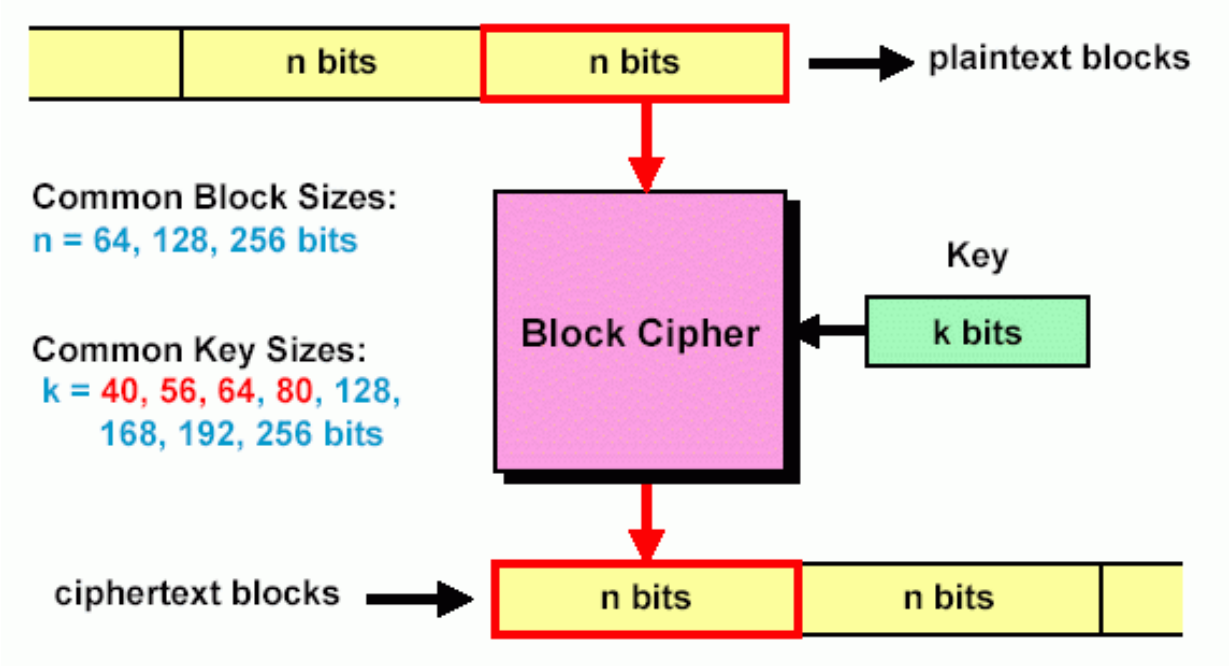
Block Cipher



Stream Cipher

Block Cipher

Divide input bit stream into n-bit sections, encrypt only that section, no dependency/history between sections



Block Cipher Principles

The design goals for block ciphers are security and efficiency.

Usually implemented in software

Many current ciphers are based on Block cipher

Most Block ciphers are based on Feistel cipher

Using the idea of product cipher

History

In the late 1960s, IBM set up a research project in computer cryptography led by Horst Feistel.

Lucifer was developed to protect the data for a cash-dispensing system that IBM had developed for Lloyds Bank in the United Kingdom.

In 1973 the US National Bureau of Standards (NBS) issued a public request for proposals for a crypto algorithm to be considered for a new cryptographic standard.

No viable submissions were received.

A second request was issued in 1974, and IBM submitted the patented Lucifer algorithm which he had devised a few years earlier.

History

The Lucifer algorithm was evaluated in secret consultations between the NBS and the U.S. National Security Agency (NSA).

After some modifications to the internal functions and a shortening of the code key size from 112 bits to 56 bits, the full details of the algorithm that was to become the Data Encryption Standard were published in the Federal Register in 1975.

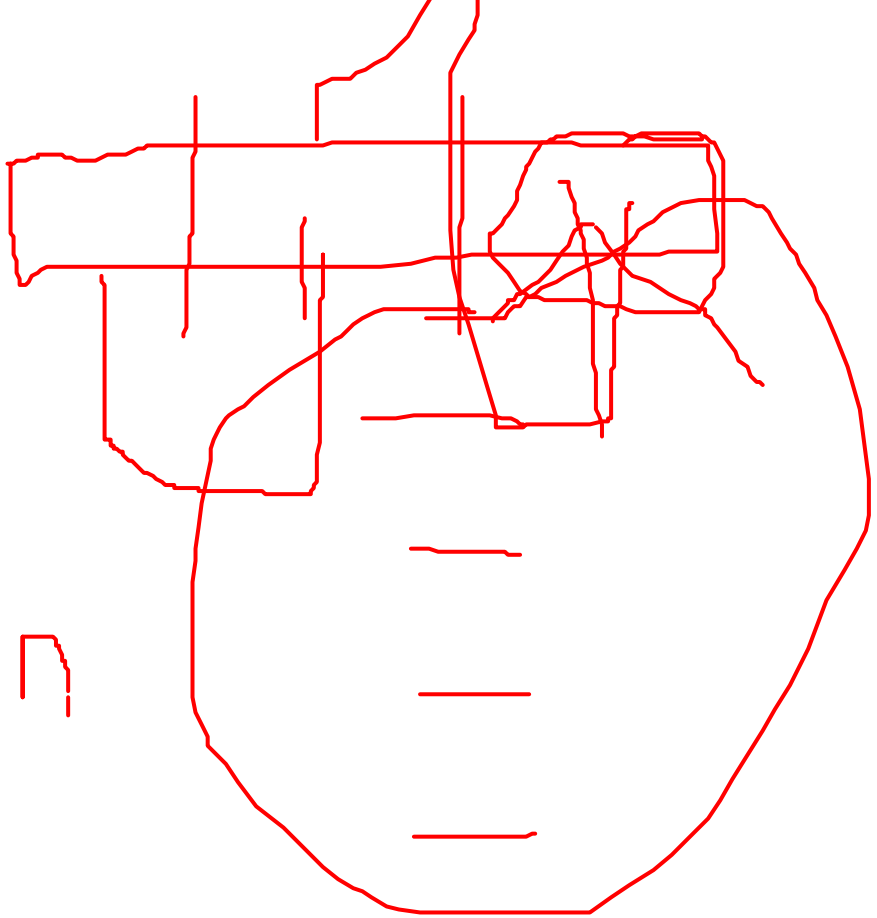
The use of the DES algorithm was made mandatory for all financial transactions of the U.S. government involving electronic fund transfer, including those conducted by member banks of the Federal Reserve System.

Subsequent adoption of the DES by standards organizations worldwide caused the DES to become a de facto international standard for business and commercial data security as well.

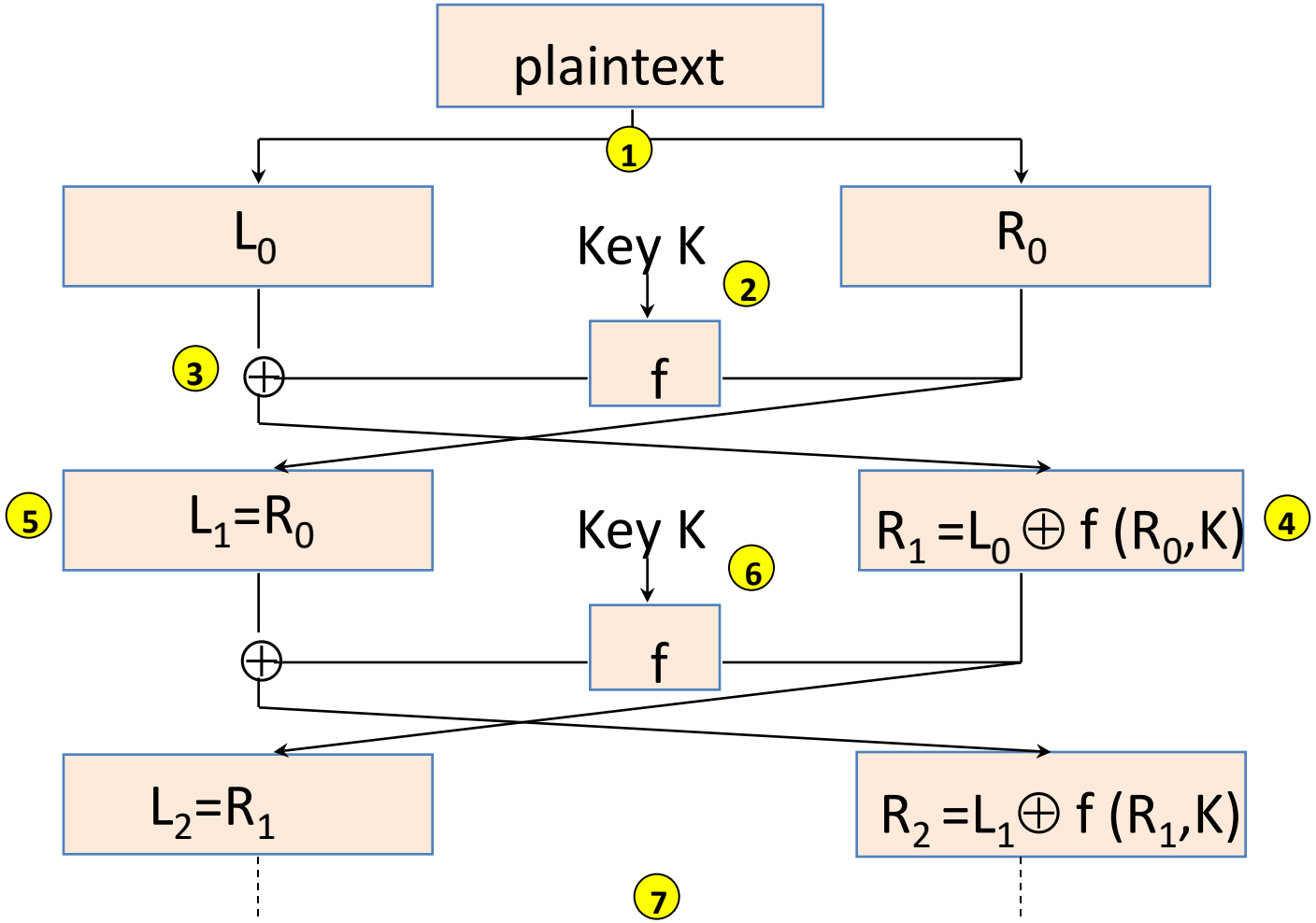
Feistel Cipher

- partitions input block into two halves
  process through multiple rounds which
  perform a substitution on left data half
  based on round function of right half & subkey
  then have permutation swapping halves

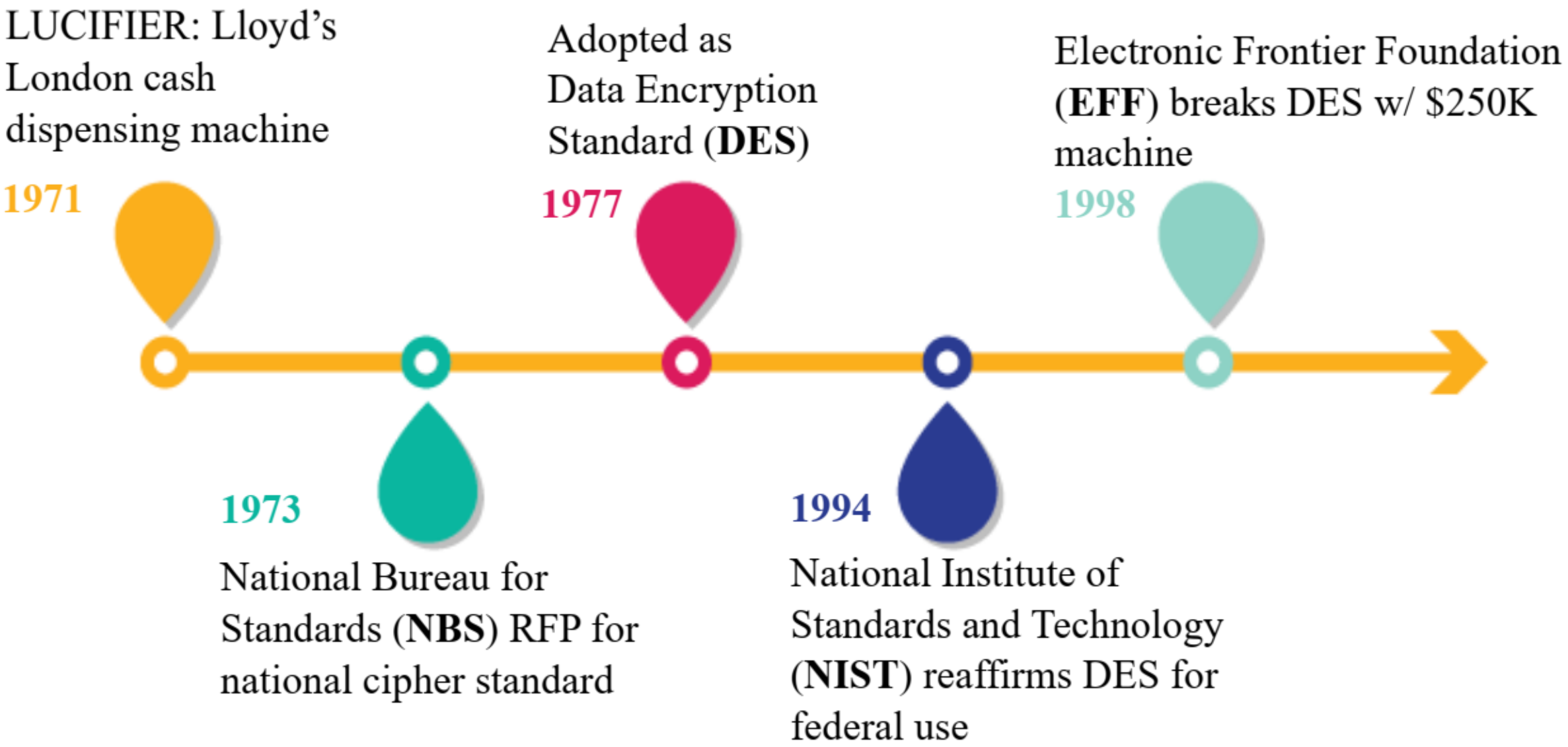
- Implements confusion and diffusion



Feistel cipher



DES


LUCIFIER: Lloyd's London cash dispensing machine
1971
1973
National Bureau for Standards (NBS) RFP for national cipher standard
Adopted as Data Encryption Standard (DES)
1977
1994
National Institute of Standards and Technology (NIST) reaffirms DES for federal use
Electronic Frontier Foundation (EFF) breaks DES w/ $250K machine
1998

DES

- The most widely used encryption scheme is based on the Data Encryption Standard (DES) adopted in 1977 by the National Bureau of Standards (NBS), now the National Institute of Standards and Technology (NIST), as Federal Information Processing Standard 46 (FIPS PUB 46).

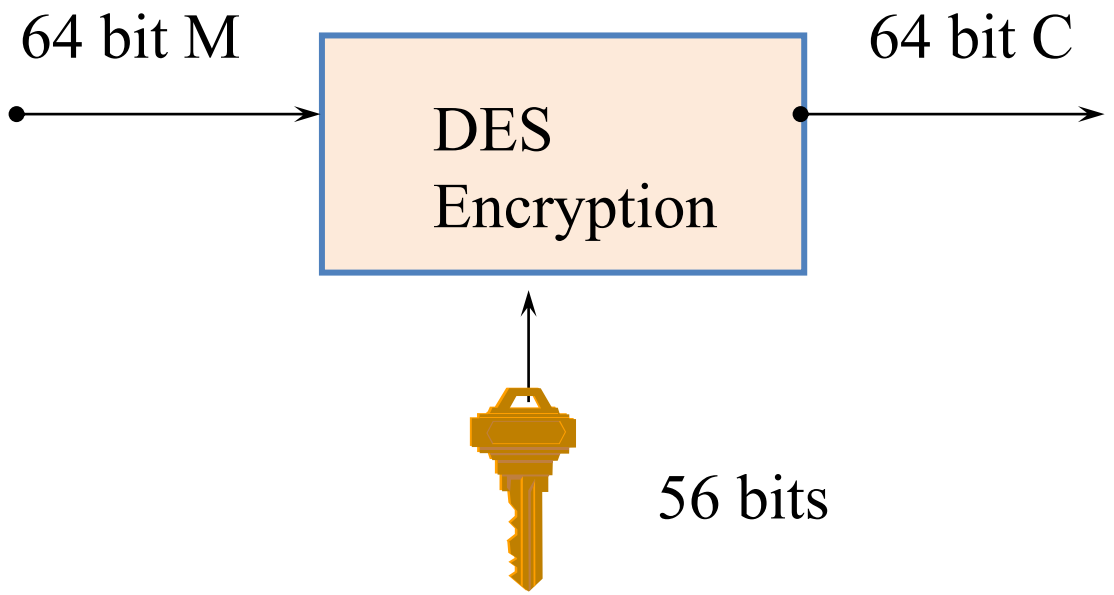
- The algorithm itself is referred to as the Data Encryption Algorithm (DEA).

- DES Characteristics:
  - Plaintext and Ciphertext – 32, 48 , 64, 128 -bit block
  - Key – 56-bit
  - Product cipher with 16 rounds
  - Follows Feistel cipher structure

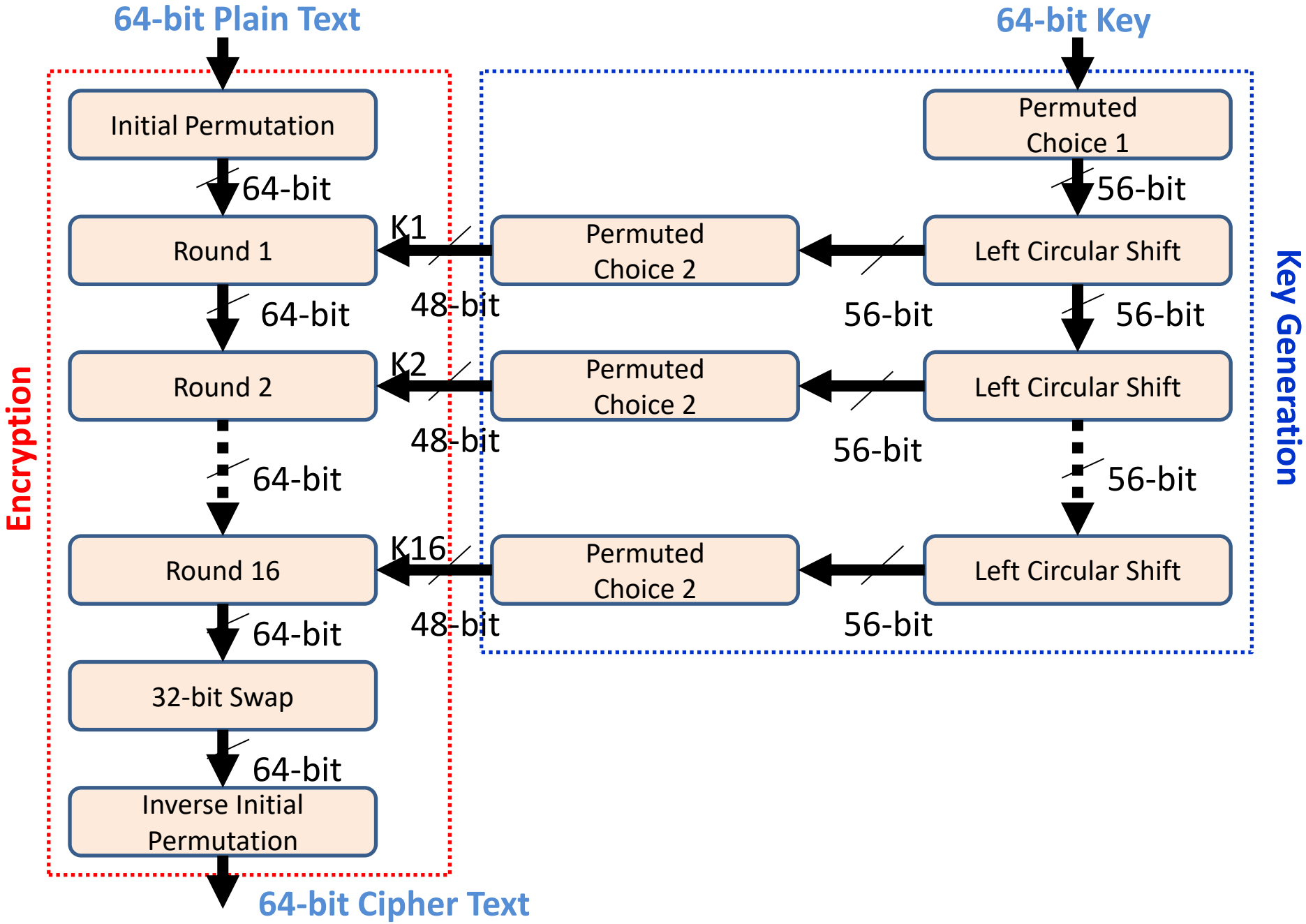
DES

Key: 64 bit quantity=8-bit parity+56-bit key Every 8th bit is a parity bit.

64 bit input, 64 bit output.



DES- Encryption



Initial Permutation

58	50	42	34	26	18	10	2
60	52	44	36	28	20	12	4
62	54	46	38	30	22	14	6
64	56	48	40	32	24	16	8
57	49	41	33	25	17	9	1
59	51	43	35	27	19	11	3
61	53	45	37	29	21	13	5
63	55	47	39	31	23	15	7

8 X 8

Inverse of Initial Permutation

40	8	48	16	56	24	64	32
39	7	47	15	55	23	63	31
38	6	46	14	54	22	62	30
37	5	45	13	53	21	61	29
36	4	44	12	52	20	60	28
35	3	43	11	51	19	59	27
34	2	42	10	50	18	58	26
33	1	41	9	49	17	57	25

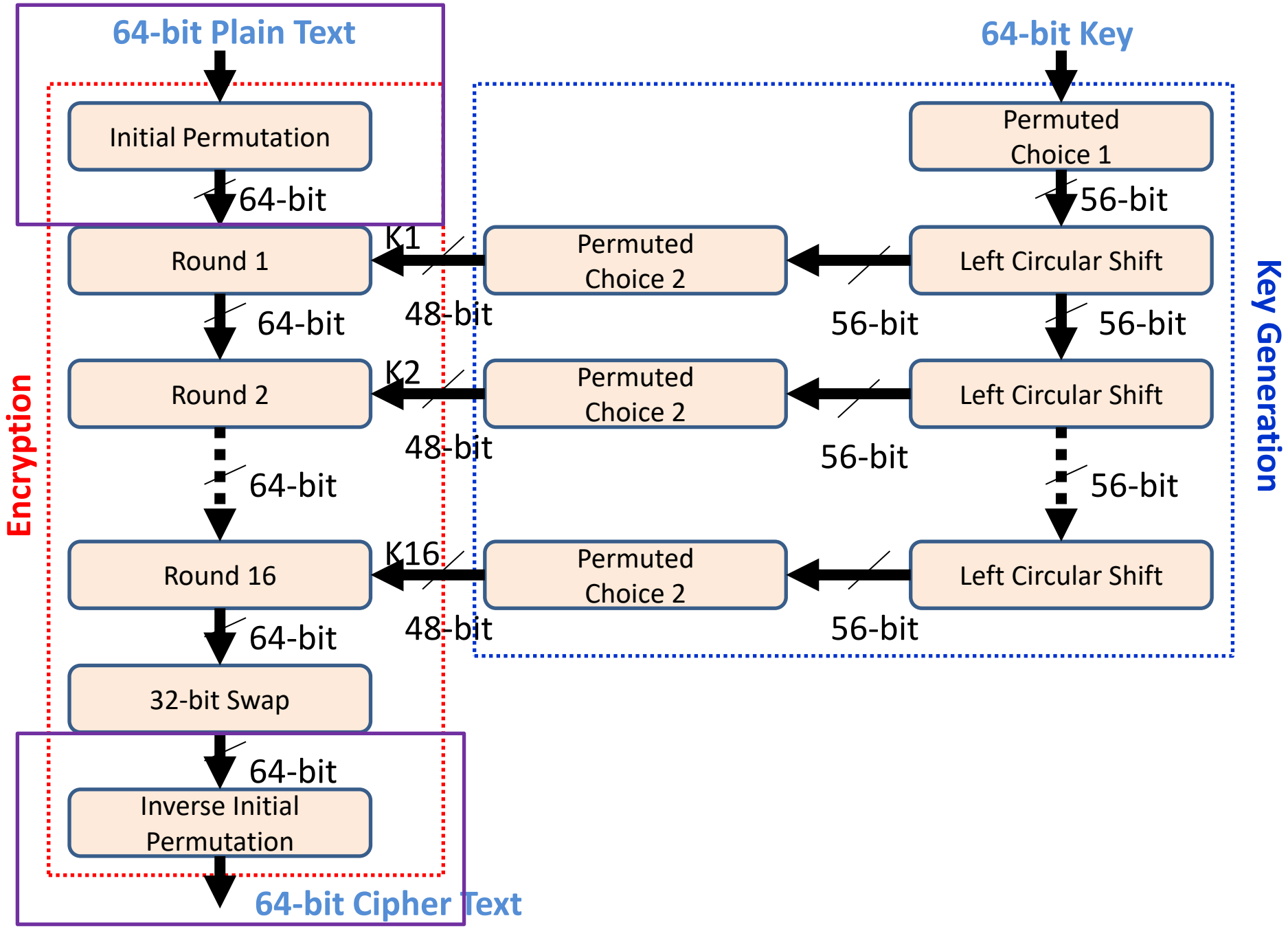
8 X 8

The meaning is as follows:

the first bit of the output is taken from the 58th bit of the input; the second bit from the 50th bit, and so on, with the last bit of the output taken from the 7th bit of the input.

This information is presented as a table for ease of presentation; it is a vector, not a matrix.

DES- Encryption



Plain Text :

0	1	1	0	0	0	0	1
0	1	1	0	0	0	1	0
0	1	1	0	0	0	1	1
0	1	1	0	0	1	0	0
0	1	1	0	0	1	0	1
0	1	1	0	0	1	1	0
0	1	1	0	0	1	1	1
0	1	1	0	1	0	0	0

1	1	1	1	1	1	1	1
0	0	0	0	0	0	0	0
0	1	1	1	1	0	0	0
0	1	0	1	0	1	0	1
0	1	0	1	0	1	0	1
0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0
0	1	1	0	0	1	1	0

Initial Permutation table:

58	50	42	34	26	18	10	2
60	52	44	36	28	20	12	4
62	54	46	38	30	22	14	6
64	56	48	40	32	24	16	8
57	49	41	33	25	17	9	1
59	51	43	35	27	19	11	3
61	53	45	37	29	21	13	5
63	55	47	39	31	23	15	7

After Permutation :

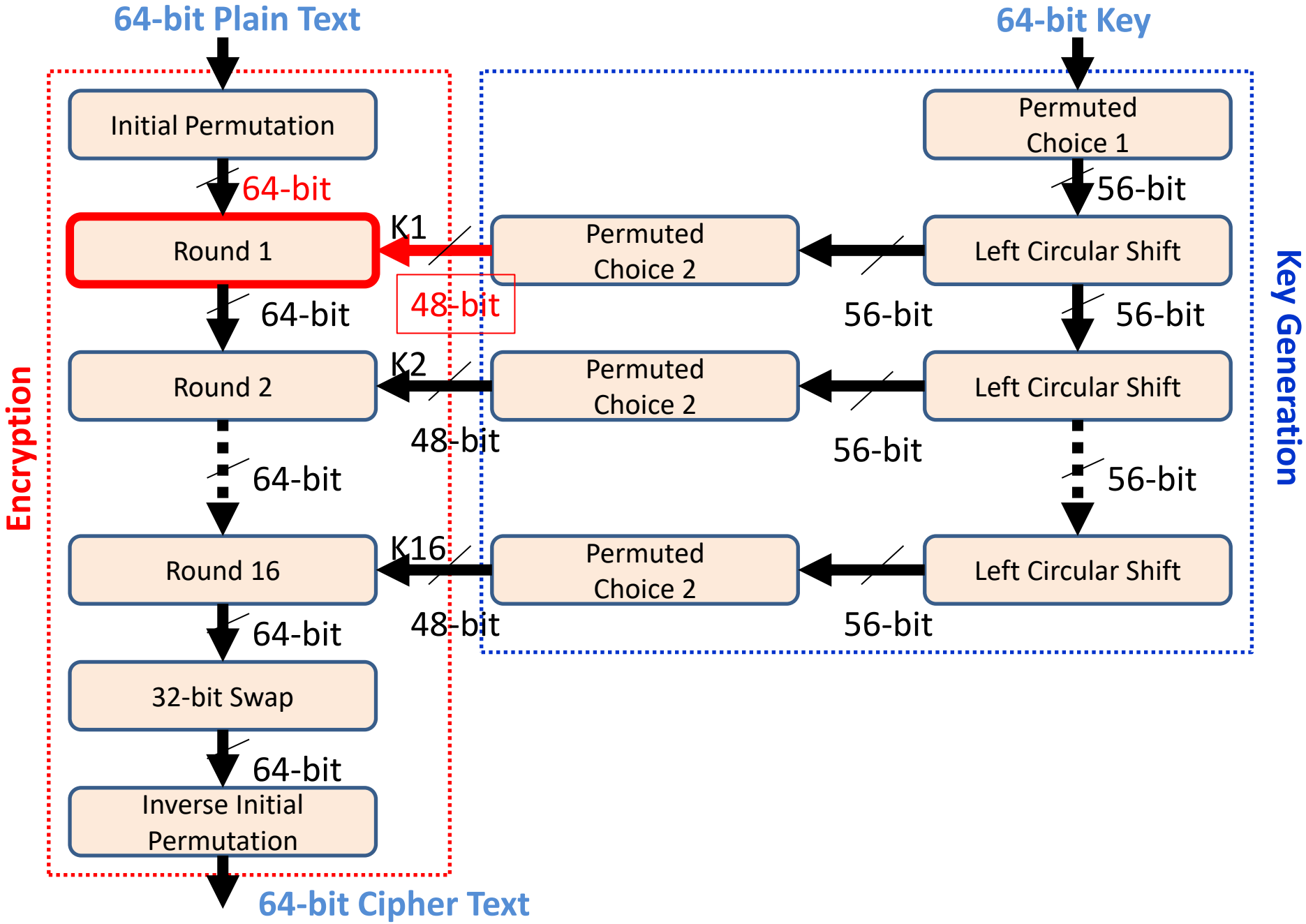
1	1	1					

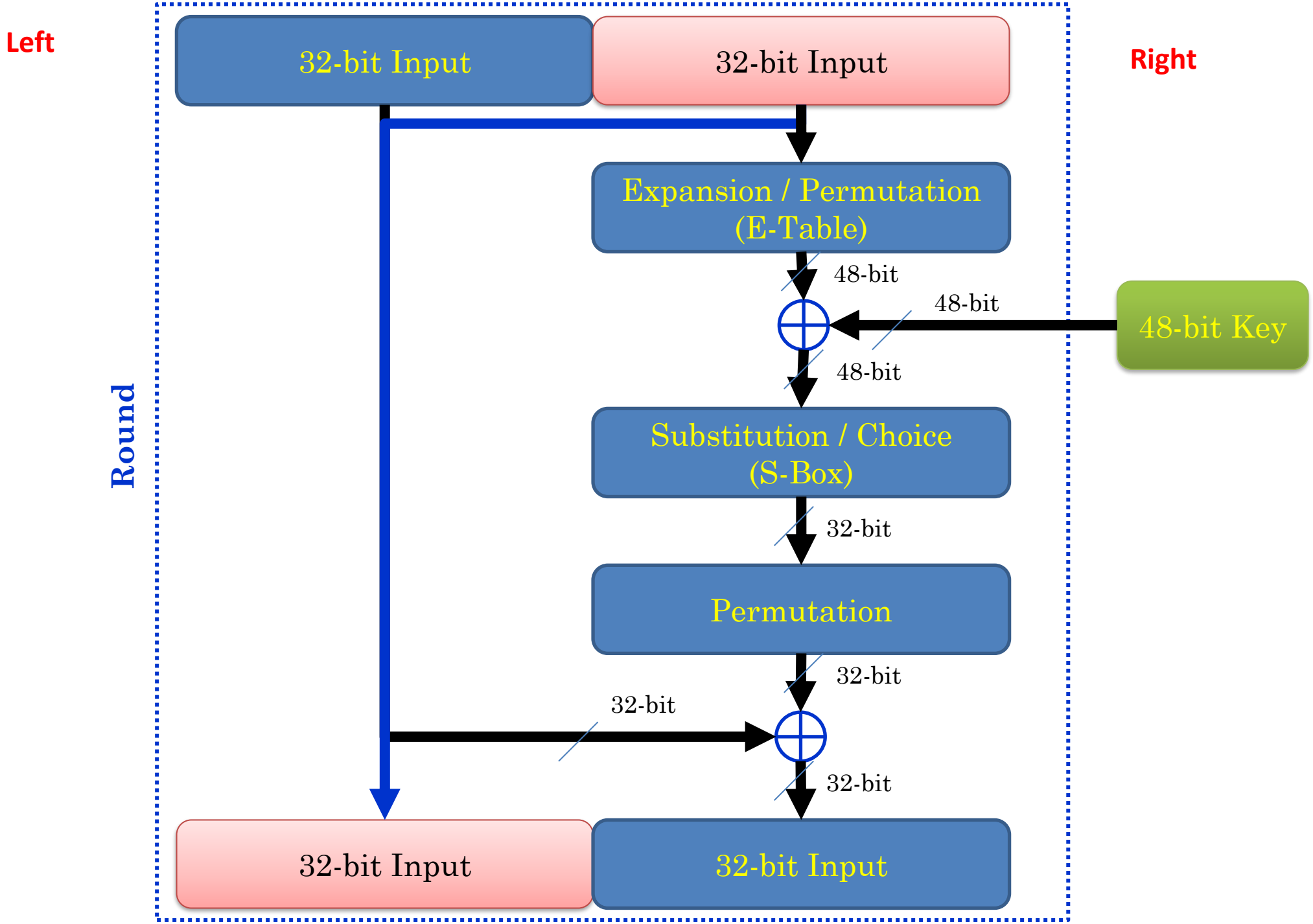
IP							
2	6	3	1	4	8	5	7

IP^{-1}							
4	1	3	5	7	2	8	6

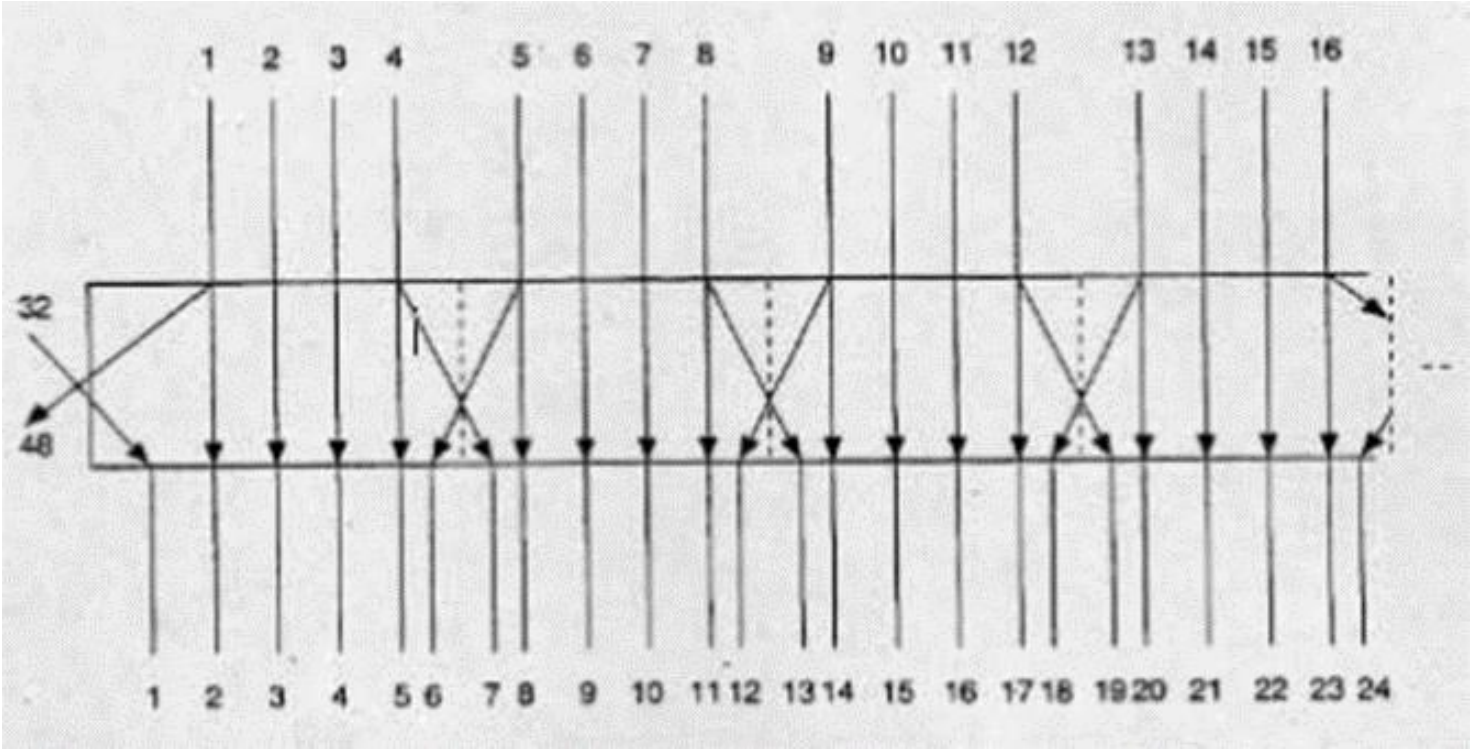


DES- Encryption




Left
Right
Round
32-bit Input
32-bit Input
Expansion / Permutation
(E-Table)
48-bit
48-bit
48-bit Key
48-bit
Substitution / Choice
(S-Box)
32-bit
Permutation
32-bit
32-bit
32-bit
32-bit Input
32-bit Input

Expansion / Permutation
(E-Table)



Input : 32 bit

You need : 48 bit

Dividing 32 bit in to four bit block (32/4 = 8 blocks)

4 bit block is replaced to 6 bit block, by adding 2 extra bits in each block ie duplicate the value

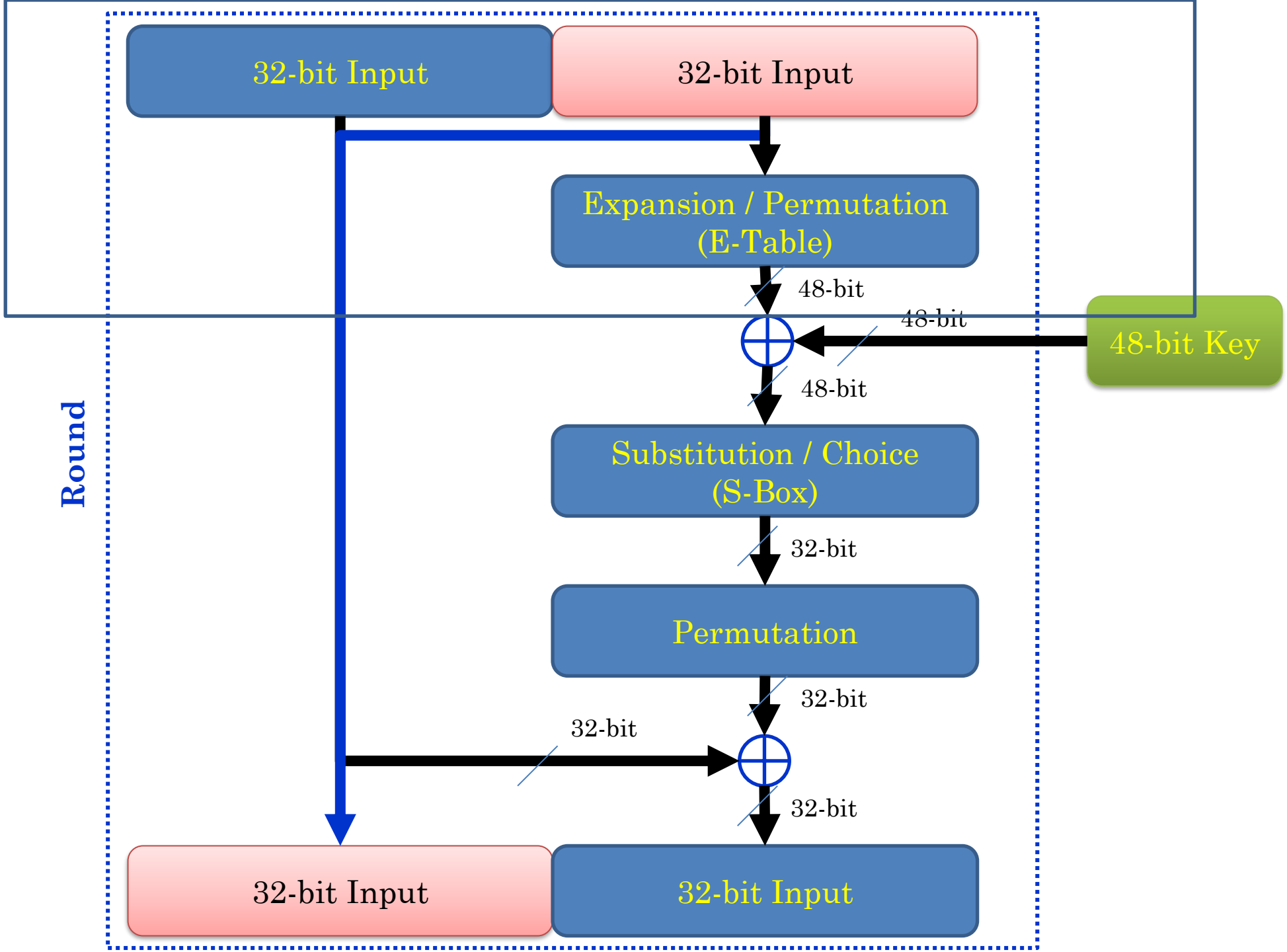
Expansion / Permutation
(E-Table)

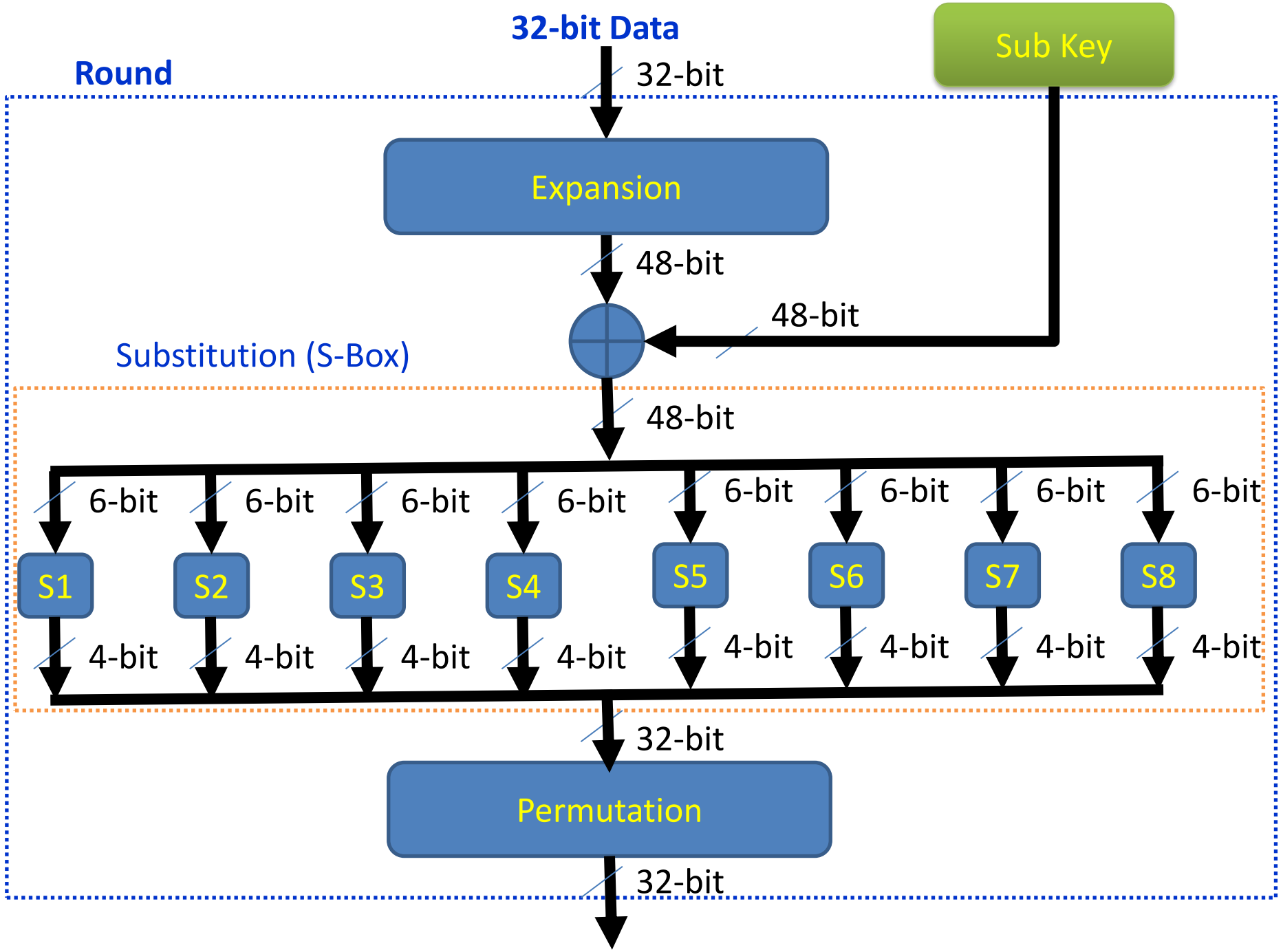
	32	1	2	3	4	5	
	4	5	6	7	8	9	
	8	9	10	11	12	13	
	12	13	14	15	16	17	
	16	17	18	19	20	21	
	20	21	22	23	24	25	
	24	25	26	27	28	29	
	28	29	30	31	32	1	

8X6

some bits from the input are duplicated at the output

e.g. the fifth bit of the input is duplicated in both the sixth and eighth bit of the output. Thus, the 32-bit half-block is expanded to 48 bits.


32-bit Input
32-bit Input
Expansion / Permutation
(E-Table)
48-bit
48-bit
48-bit Key
48-bit
Substitution / Choice
(S-Box)
32-bit
Permutation
32-bit
32-bit
32-bit
32-bit Input
32-bit Input
Round


32-bit Data
Sub Key
Round
32-bit
Expansion
48-bit
48-bit
Substitution (S-Box)
48-bit
6-bit
6-bit
6-bit
6-bit
6-bit
6-bit
6-bit
6-bit
S1
S2
S3
S4
S5
S6
S7
S8
4-bit
4-bit
4-bit
4-bit
4-bit
4-bit
4-bit
4-bit
32-bit
Permutation
32-bit

S-Box

S_1

14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
0	15	7	4	14	2	13	1	10	6	12	11	9	5	3	8
4	1	14	8	13	6	2	11	15	12	9	7	3	10	5	0
15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13

S_2

15	1	8	14	6	11	3	4	9	7	2	13	12	0	5	10
3	13	4	7	15	2	8	14	12	0	1	10	6	9	11	5
0	14	7	11	10	4	13	1	5	8	12	6	9	3	2	15
13	8	10	1	3	15	4	2	11	6	7	12	0	5	14	9

S_3

10	0	9	14	6	3	15	5	1	13	12	7	11	4	2	8
13	7	0	9	3	4	6	10	2	8	5	14	12	11	15	1
13	6	4	9	8	15	3	0	11	1	2	12	5	10	14	7
1	10	13	0	6	9	8	7	4	15	14	3	11	5	2	12

S_4

7	13	14	3	0	6	9	10	1	2	8	5	11	12	4	15
13	8	11	5	6	15	0	3	4	7	2	12	1	10	14	9
10	6	9	0	12	11	7	13	15	1	3	14	5	2	8	4
3	15	0	6	10	1	13	8	9	4	5	11	12	7	2	14

S_5

2	12	4	1	7	10	11	6	8	5	3	15	13	0	14	9
14	11	2	12	4	7	13	1	5	0	15	10	3	9	8	6
4	2	1	11	10	13	7	8	15	9	12	5	6	3	0	14
11	8	12	7	1	14	2	13	6	15	0	9	10	4	5	3

S_6

12	1	10	15	9	2	6	8	0	13	3	4	14	7	5	11
10	15	4	2	7	12	9	5	6	1	13	14	0	11	3	8
9	14	15	5	2	8	12	3	7	0	4	10	1	13	11	6
4	3	2	12	9	5	15	10	11	14	1	7	6	0	8	13

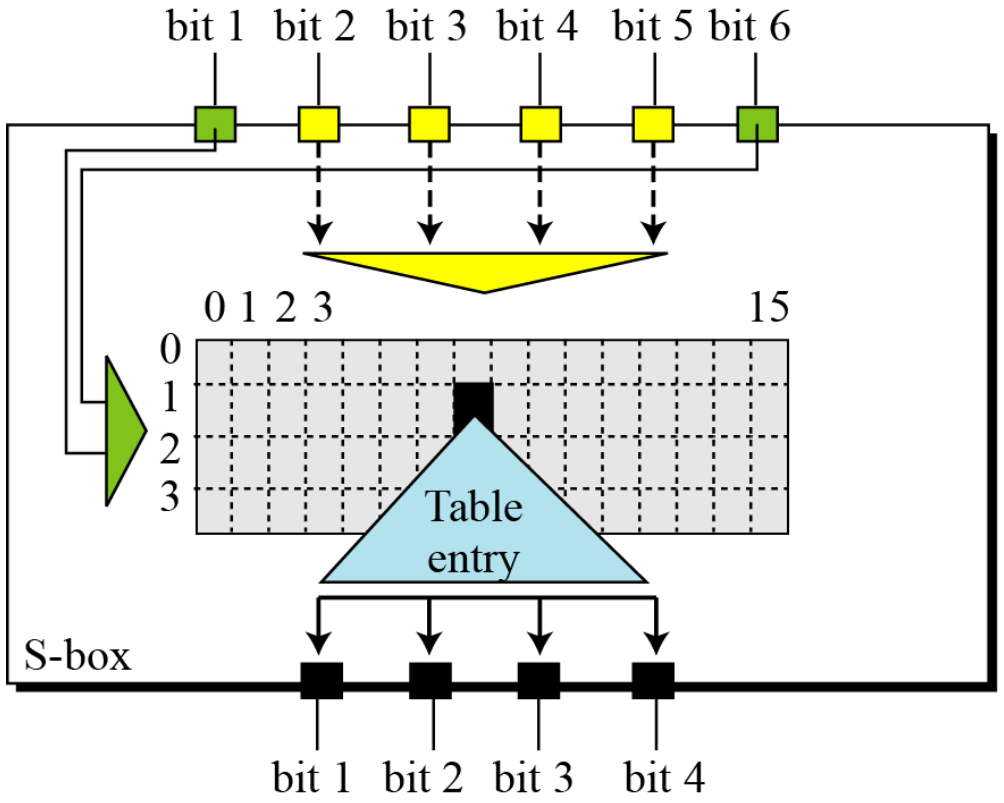
S_7

4	11	2	14	15	0	8	13	3	12	9	7	5	10	6	1
13	0	11	7	4	9	1	10	14	3	5	12	2	15	8	6
1	4	11	13	12	3	7	14	10	15	6	8	0	5	9	2
6	11	13	8	1	4	10	7	9	5	0	15	14	2	3	12

S_8

13	2	8	4	6	15	11	1	10	9	3	14	5	0	12	7
1	15	13	8	10	3	7	4	12	5	6	11	0	14	9	2
7	11	4	1	9	12	14	2	0	6	10	13	15	3	5	8
2	1	14	7	4	10	8	13	15	12	9	0	3	5	6	11

S-box rule



Given a 6-bit input, the 4-bit output is found by selecting the row using the outer two bits, and the column using the inner four bits.

permutation for S-box 1.

S-box 1

	0	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*	*12*	*13*	*14*	*15*
0	14	04	13	01	02	15	11	08	03	10	06	12	05	09	00	07
1	00	15	07	04	14	02	13	10	03	06	12	11	09	05	03	08
2	04	01	14	08	13	06	02	11	15	12	09	07	03	10	05	00
3	15	12	08	02	04	09	01	07	05	11	03	14	10	00	06	13

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	14	04	13	01	02	15	11	08	03	10	06	12	05	09	00	07
1	00	15	07	04	14	02	13	10	03	06	12	11	09	05	03	08
2	04	01	14	08	13	06	02	11	15	12	09	07	03	10	05	00
3	15	12	08	02	04	09	01	07	05	11	03	14	10	00	06	13

The remaining bits are 0001 in binary, which is 1 in decimal. We look for the value in row 3, column 1, in S-box 1.

The result is 12 in decimal, which in binary is 1100.

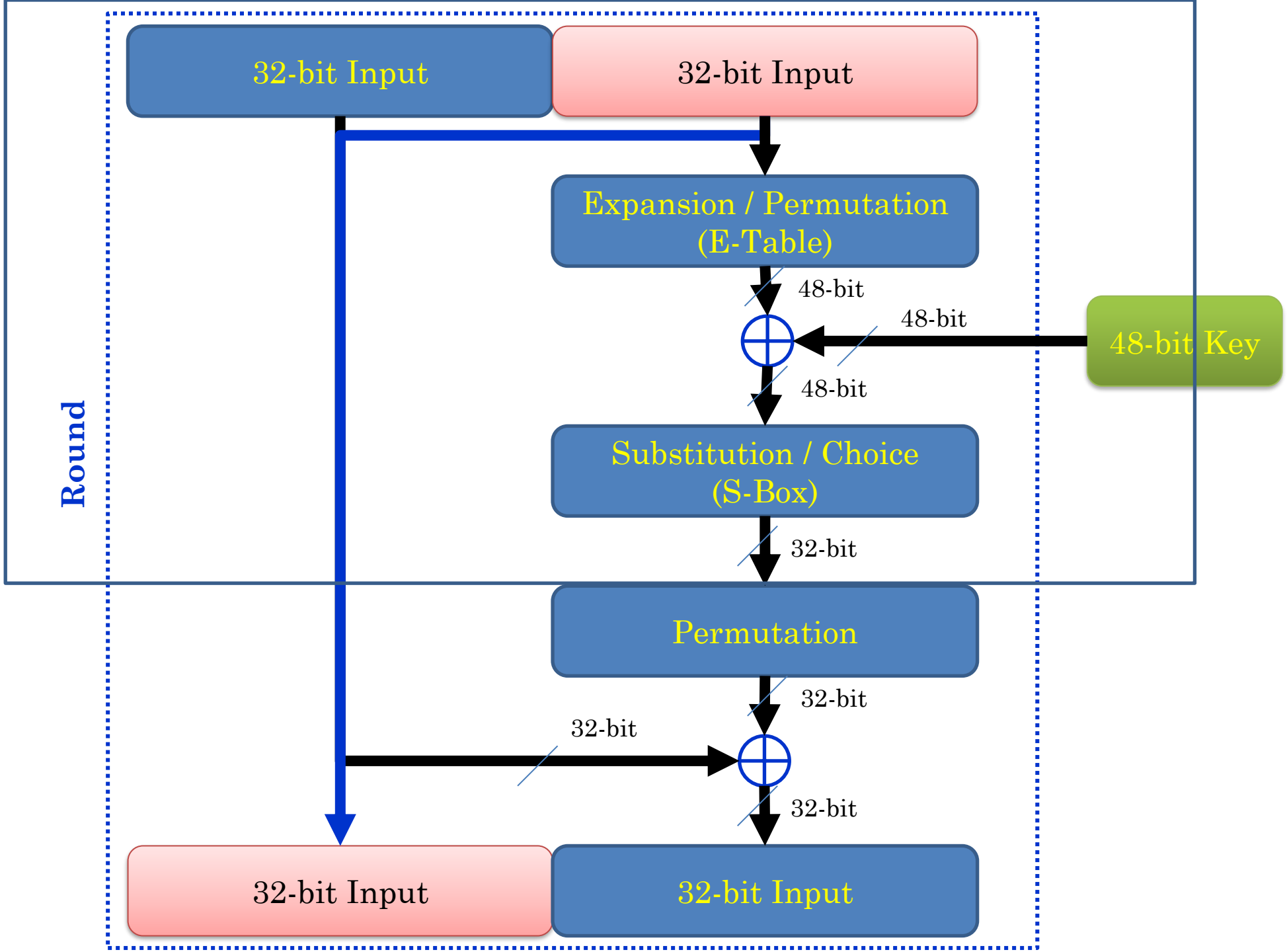
So the input 100011 yields the output 1100.

TRY

	0	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*	*12*	*13*	*14*	*15*
0	14	04	13	01	02	15	11	08	03	10	06	12	05	09	00	07
1	00	15	07	04	14	02	13	10	03	06	12	11	09	05	03	08
2	04	01	14	08	13	06	02	11	15	12	09	07	03	10	05	00
3	15	12	08	02	04	09	01	07	05	11	03	14	10	00	06	13

The input for s box is given below find the output

011010 , 110010 , 000011


Round
32-bit Input
32-bit Input
Expansion / Permutation
(E-Table)
48-bit
48-bit
48-bit Key
48-bit
Substitution / Choice
(S-Box)
32-bit
Permutation
32-bit
32-bit
32-bit
32-bit Input
32-bit Input

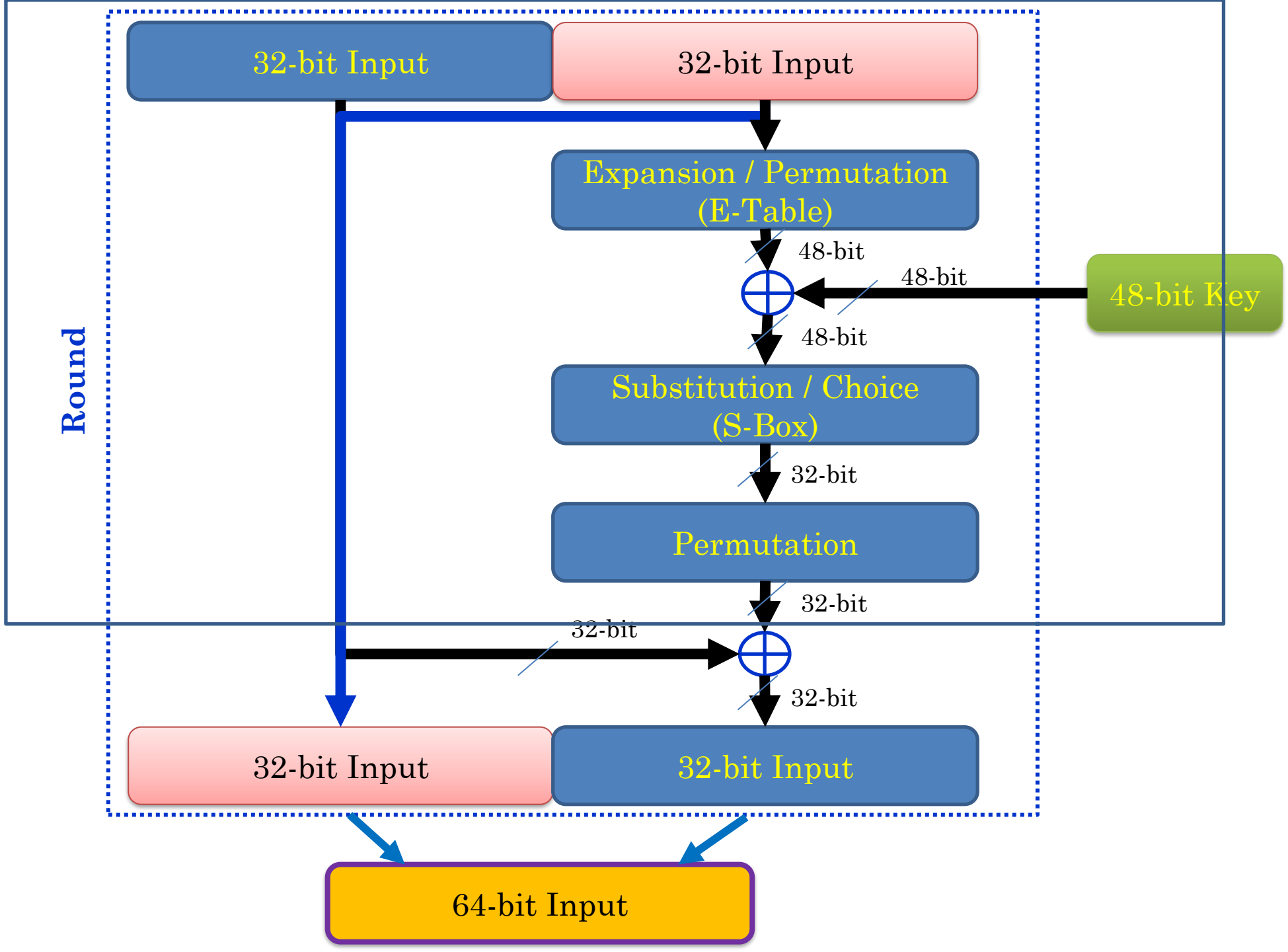
P- box Permutation

16	7	20	21	29	12	28	17
1	15	23	26	5	18	31	10
2	8	24	14	32	27	3	9
19	13	30	6	22	11	4	25

The 32-bit output of the s box is permuted to a P box

This permutation maps each input bit to an output position, no bits are used to twice and no bits are ignored.

The second bit is moved to the 7th position , the third bit is moved to 20th position , so on …
This is called straight permutation


Round
32-bit Input
32-bit Input
Expansion / Permutation
(E-Table)
48-bit
48-bit
48-bit Key
48-bit
Substitution / Choice
(S-Box)
32-bit
Permutation
32-bit
32-bit
32-bit
32-bit Input
32-bit Input
64-bit Input

Second XOR Operation

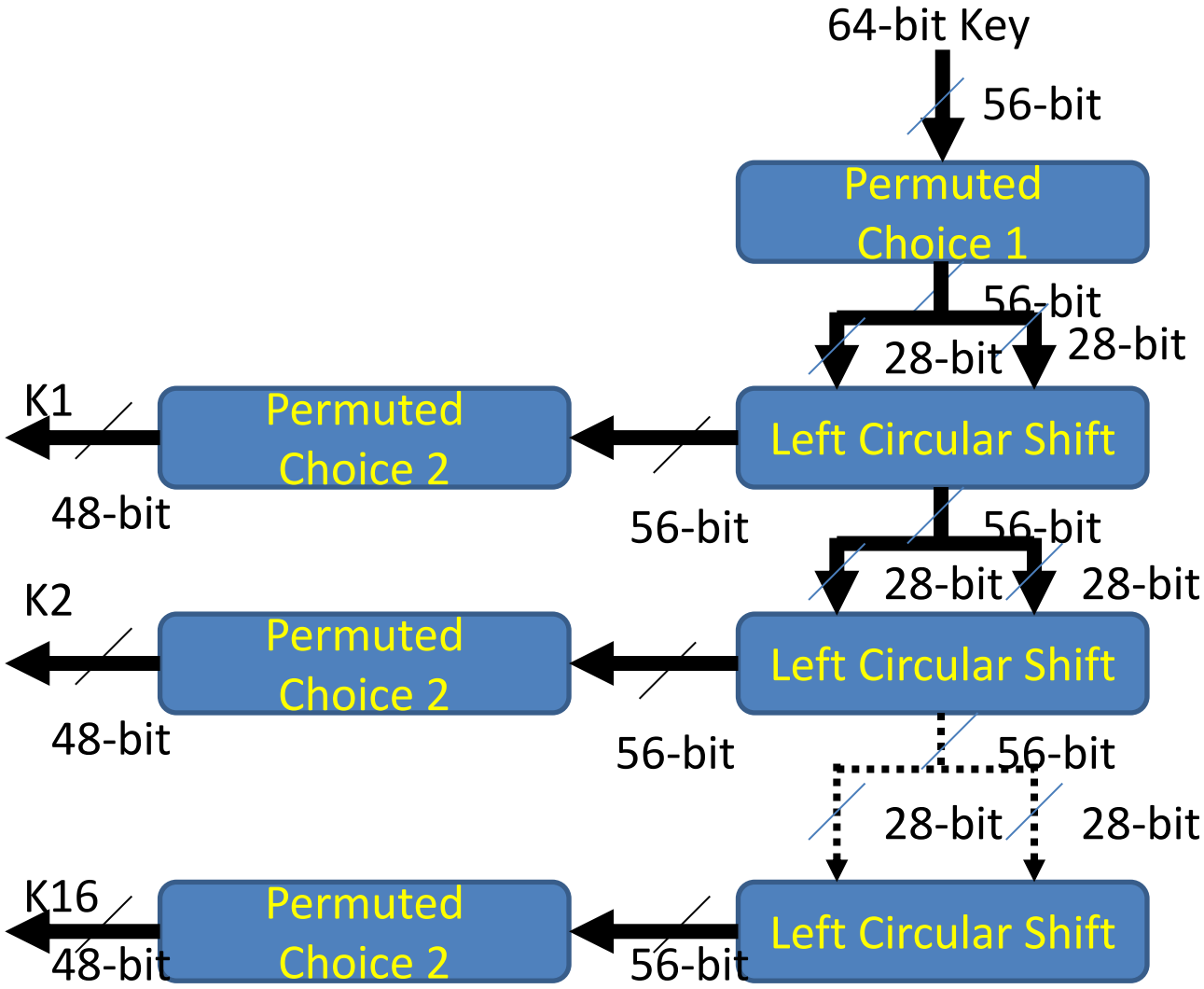
- Output of P-box is XORed with the left half of 64-bit input block

```
      10001101111010110010101100101111
XOR   11100010101111010001110001100 1101
      ————————————————————————————————
      01101111011011000110111010010010
```

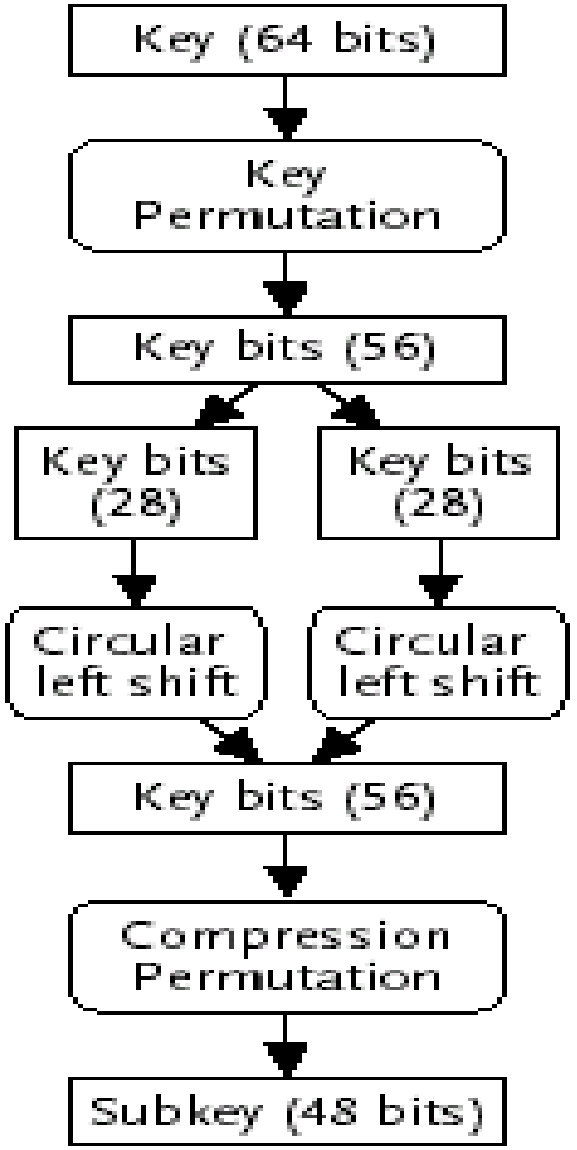
- 32-bit output of the XOR operation:

  – 01101111011011000110111010010010

Key Generation



Key Generation



Key Generation

Initially 64 bit key is reduced to a 56 bit key by ignoring every eighth bit.

These bits are used as parity check to ensure the key is error-free.

After 56-bit key is extracted, a different 48 bit sub key is generated for each 16 rounds.

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56
57	58	59	60	61	62	63	64

Key Permutation

57	49	41	33	25	17	9
1	58	50	42	34	26	18
10	2	59	51	43	35	27
19	11	3	60	52	44	36
63	55	47	39	31	23	15
7	62	54	46	38	30	22
14	6	61	53	45	37	29
21	13	5	28	20	12	4

Key Generation

56 bit is divided in to two halves

28 bit 28 bit

Each half is subject to a cyclic shift of 1 or 2 bits in each round as follows (total 16 rounds)

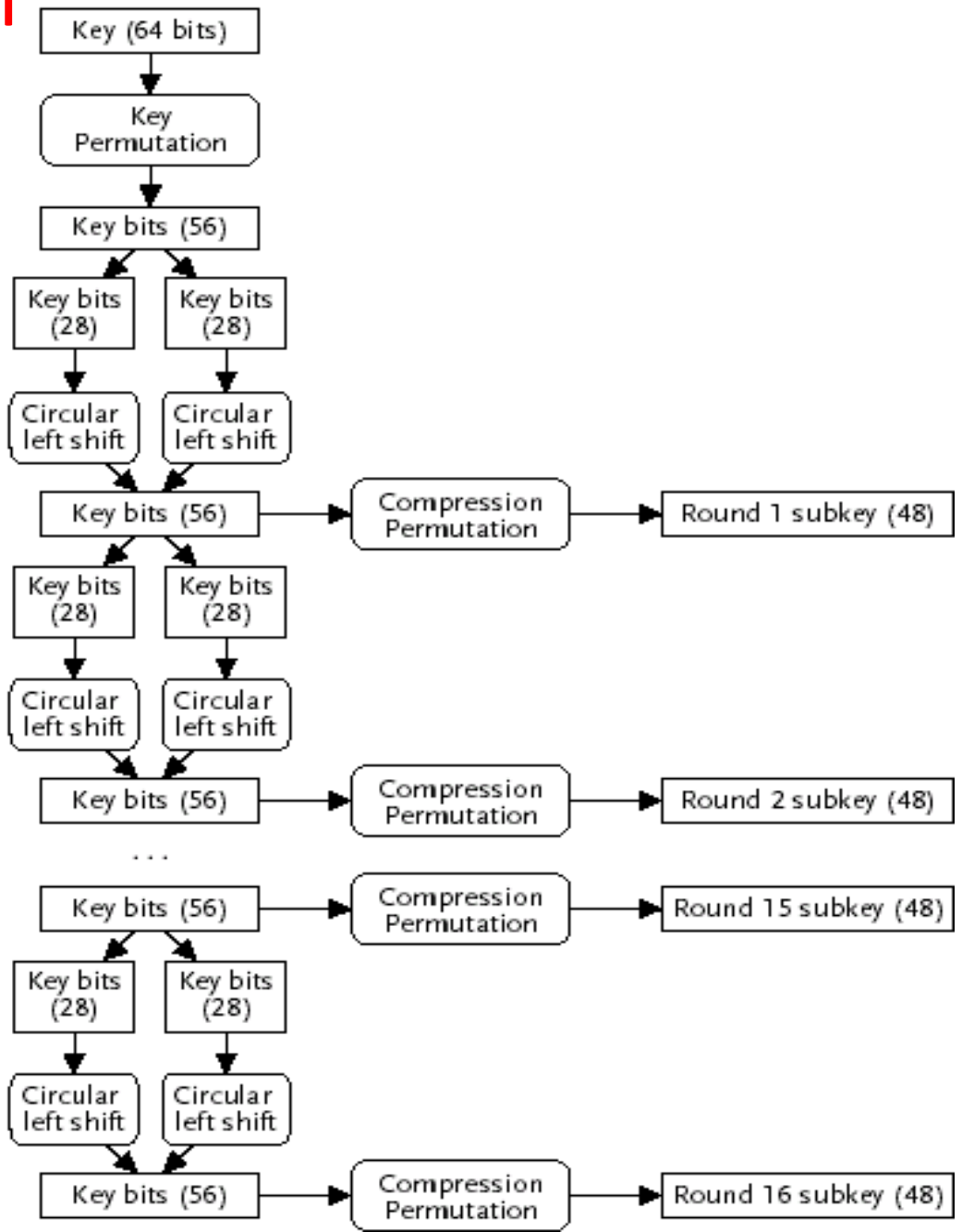
Round Number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Bits Rotated	1	1	2	2	2	2	2	2	1	2	2	2	2	2	2	1

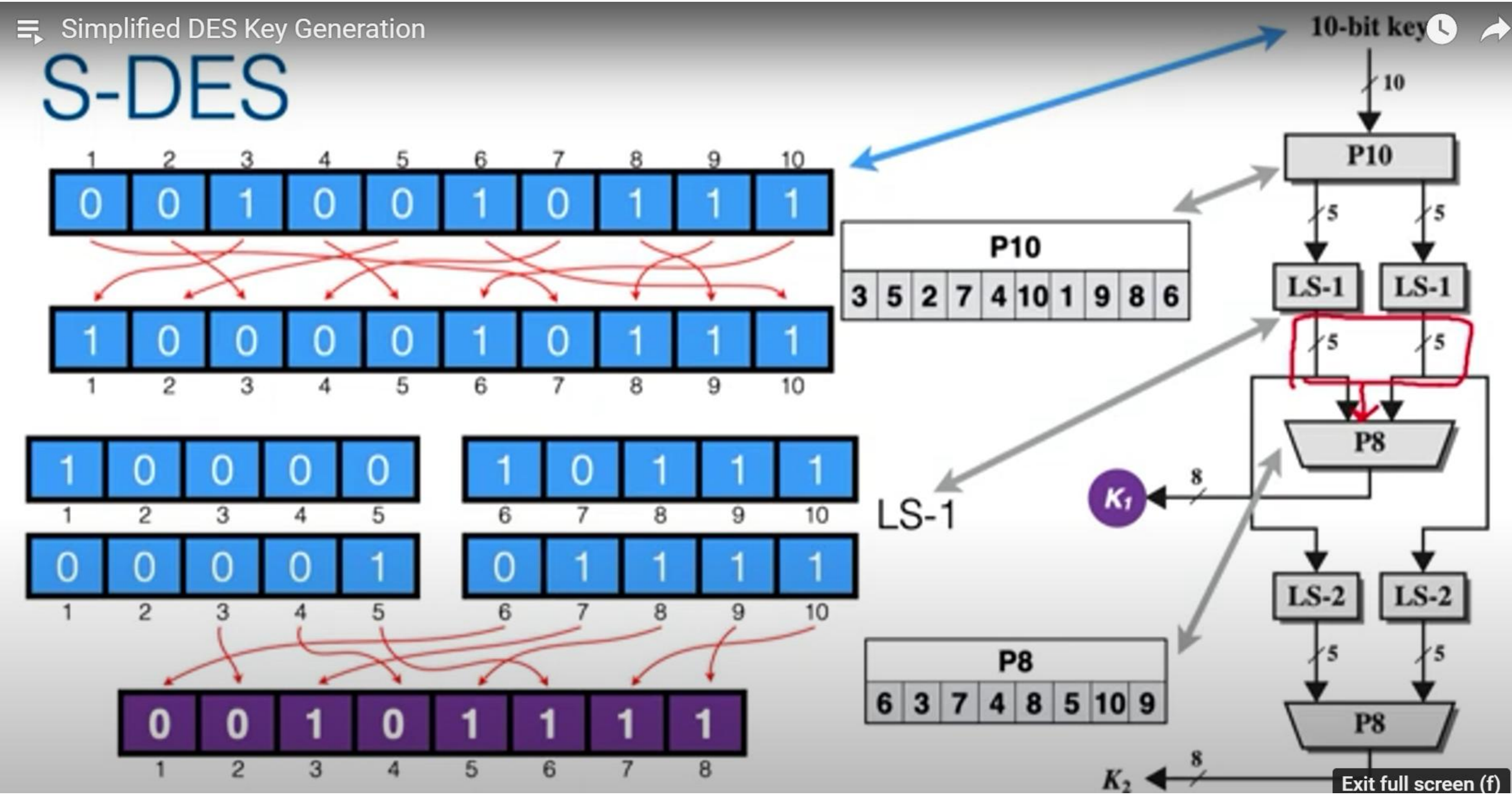
Compression permutation table

14	17	11	24	1	5	3	28
15	6	21	10	23	19	12	4
26	8	16	7	27	20	13	2
41	52	31	37	47	55	30	40
51	45	33	48	44	49	39	56
34	53	46	42	50	36	29	32

The bit in the position 33 of the shifted key moves to position 35 of the output, and the bit in position 18 of the shifted key is ignored.

Key Generation – overall




Simplified DES Key Generation
S-DES
1 2 3 4 5 6 7 8 9 10
0 0 1 0 0 1 0 1 1 1
1 0 0 0 0 1 0 1 1 1
1 2 3 4 5 6 7 8 9 10
1 0 0 0 0
1 2 3 4 5
1 0 1 1 1
6 7 8 9 10
0 0 0 0 1
1 2 3 4 5
0 1 1 1 1
6 7 8 9 10
0 0 1 0 1 1 1 1
1 2 3 4 5 6 7 8
10-bit key
10
P10
5 5
LS-1 LS-1
5 5
P8
8
K1
LS-2 LS-2
5 5
P8
8
K2
P10
3 5 2 7 4 10 1 9 8 6
LS-1
P8
6 3 7 4 8 5 10 9
Exit full screen (f)

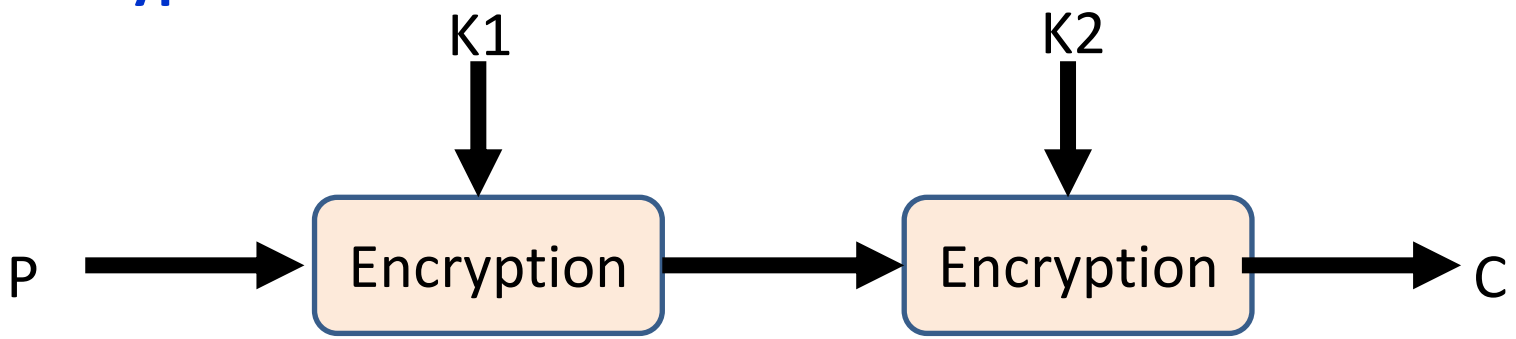
Avalanche effect

A desirable property of any encryption algorithm is that a small change in either plaintext or key should produce significant changes in the ciphertext.

DES exhibits a strong avalanche effect.

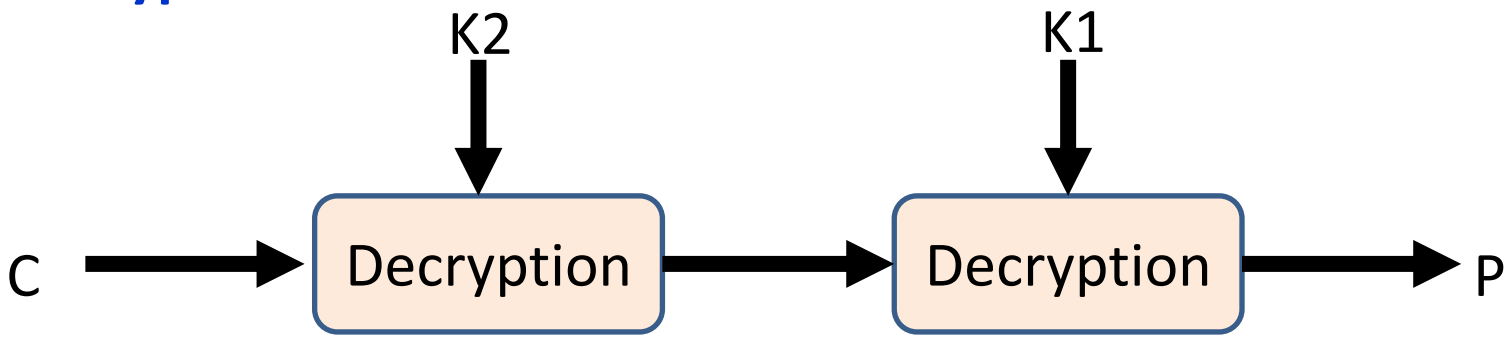
Double DES

(a) Encryption



C = E($K2$, E($K1$, P))

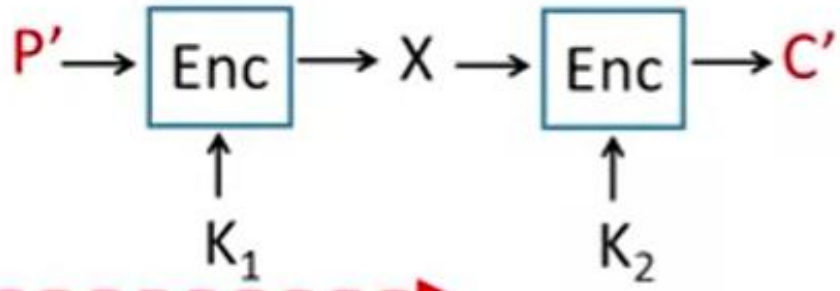
(b) Decryption



P = D($K1$, D($K2$, C))

Meet in the Middle Attack





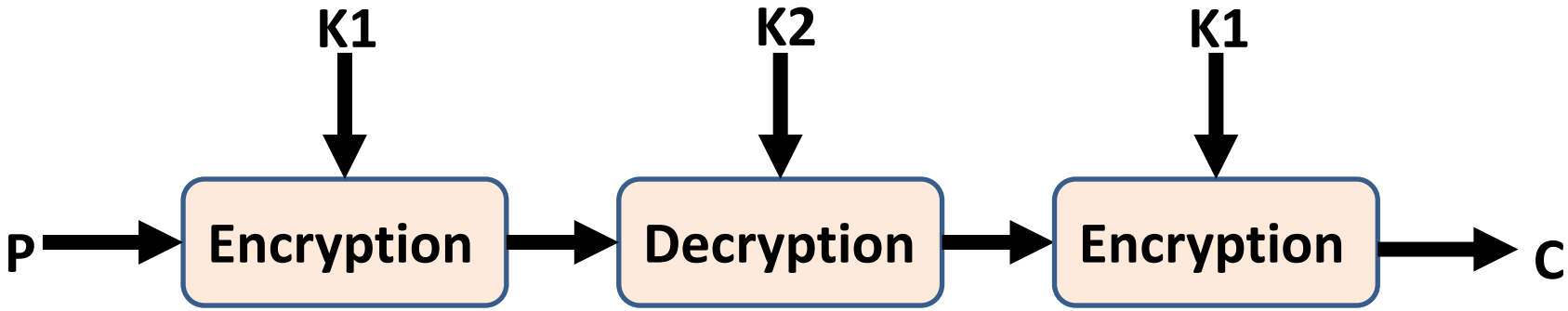
Compute and store
2^{56} P'→X mappings
using different K_1's

Compute 2^{56} C'→X
decryptions using K_2's

Compare X's from two directions;
If the same, try with different
known plaintexts (P'',C'')

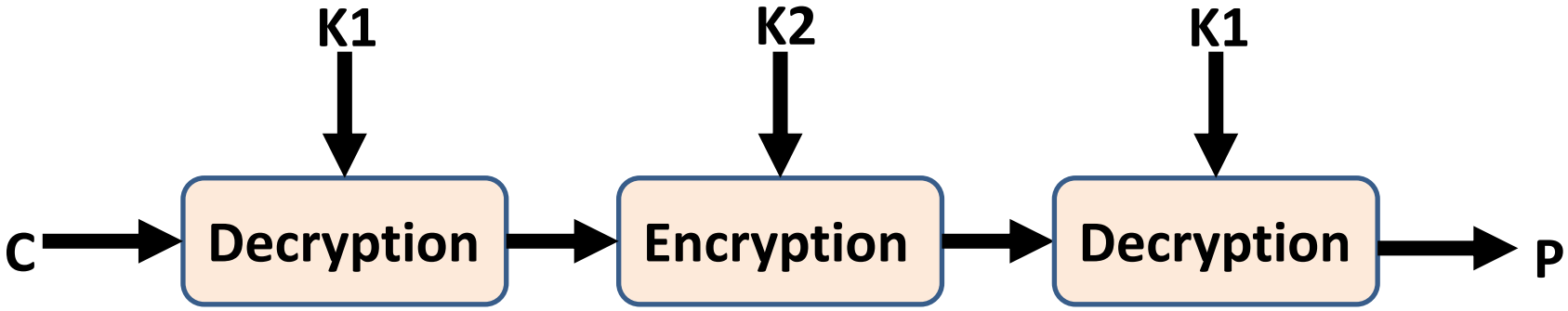
Triple DES with Two Keys

(a) Encryption



C = E($K1$, D($K2$, E($K1$, P)))

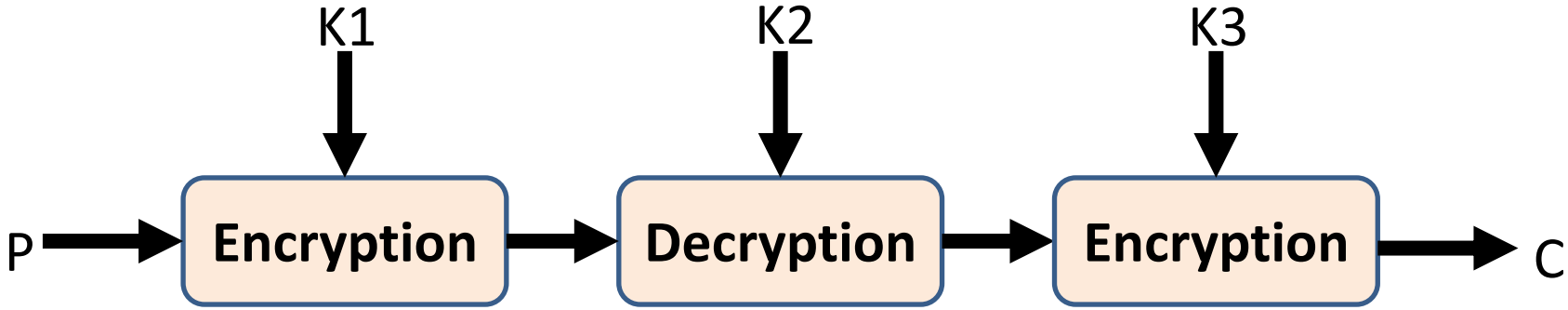
(b) Decryption



P = D($K1$, E($K2$, D($K1$, C)))

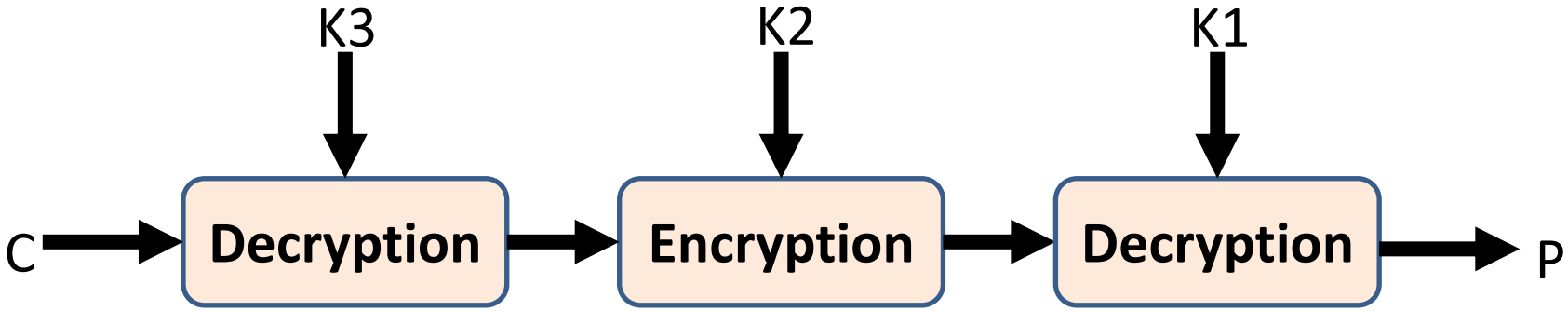
Triple DES with Three Keys

(a) Encryption



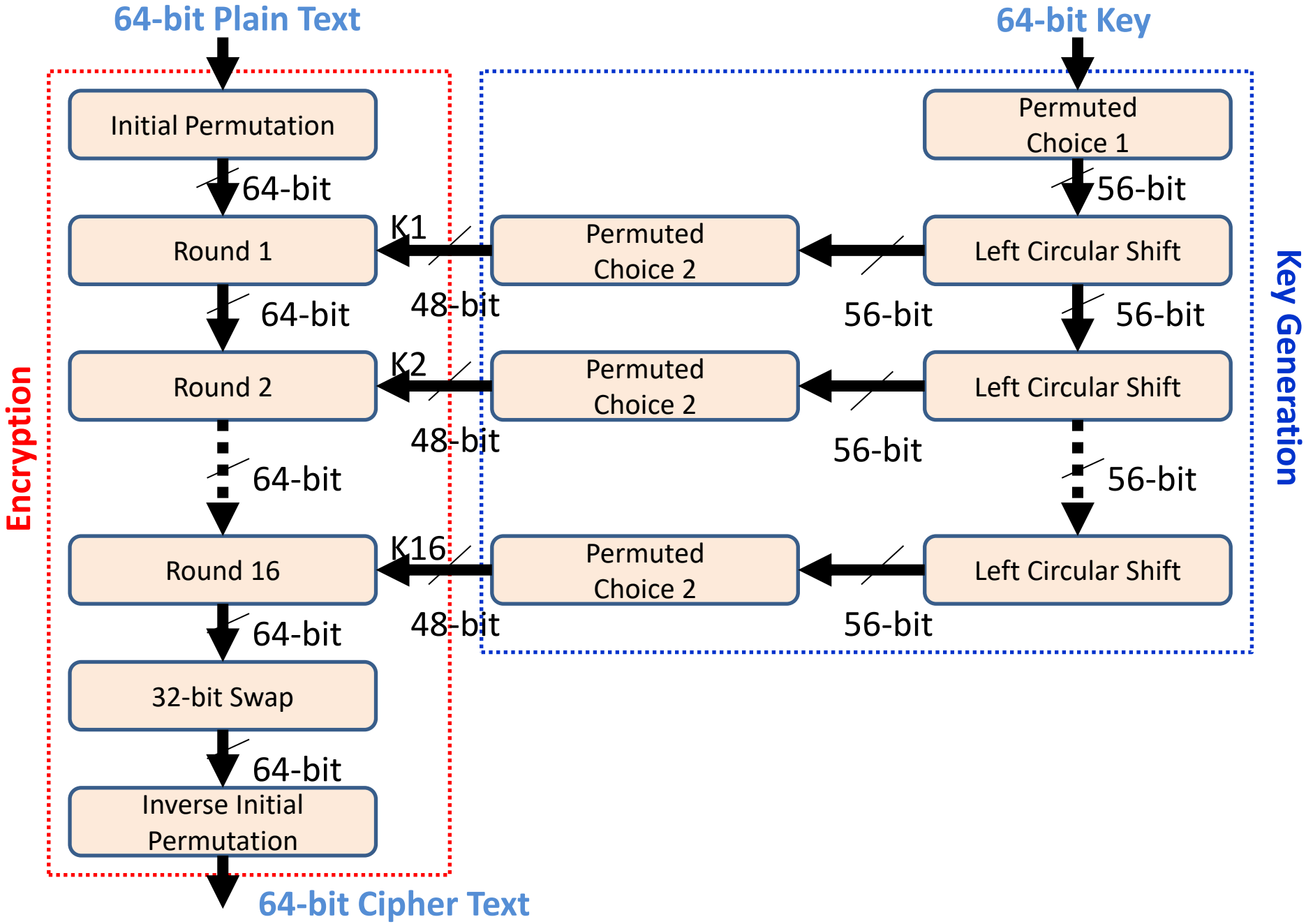
C = E($K3$, D($K2$, E($K1$, P)))

(b) Decryption



P = D($K1$, E($K2$, D($K3$, C)))

DES- Encryption



Plain Text :

0	1	1	0	0	0	0	1
0	1	1	0	0	0	1	0
0	1	1	0	0	0	1	1
0	1	1	0	0	1	0	0
0	1	1	0	0	1	0	1
0	1	1	0	0	1	1	0
0	1	1	0	0	1	1	1
0	1	1	0	1	0	0	0

1	1	1	1	1	1	1	1
0	0	0	0	0	0	0	0
0	1	1	1	1	0	0	0
0	1	0	1	0	1	0	1
0	1	0	1	0	1	0	1
0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0
0	1	1	0	0	1	1	0

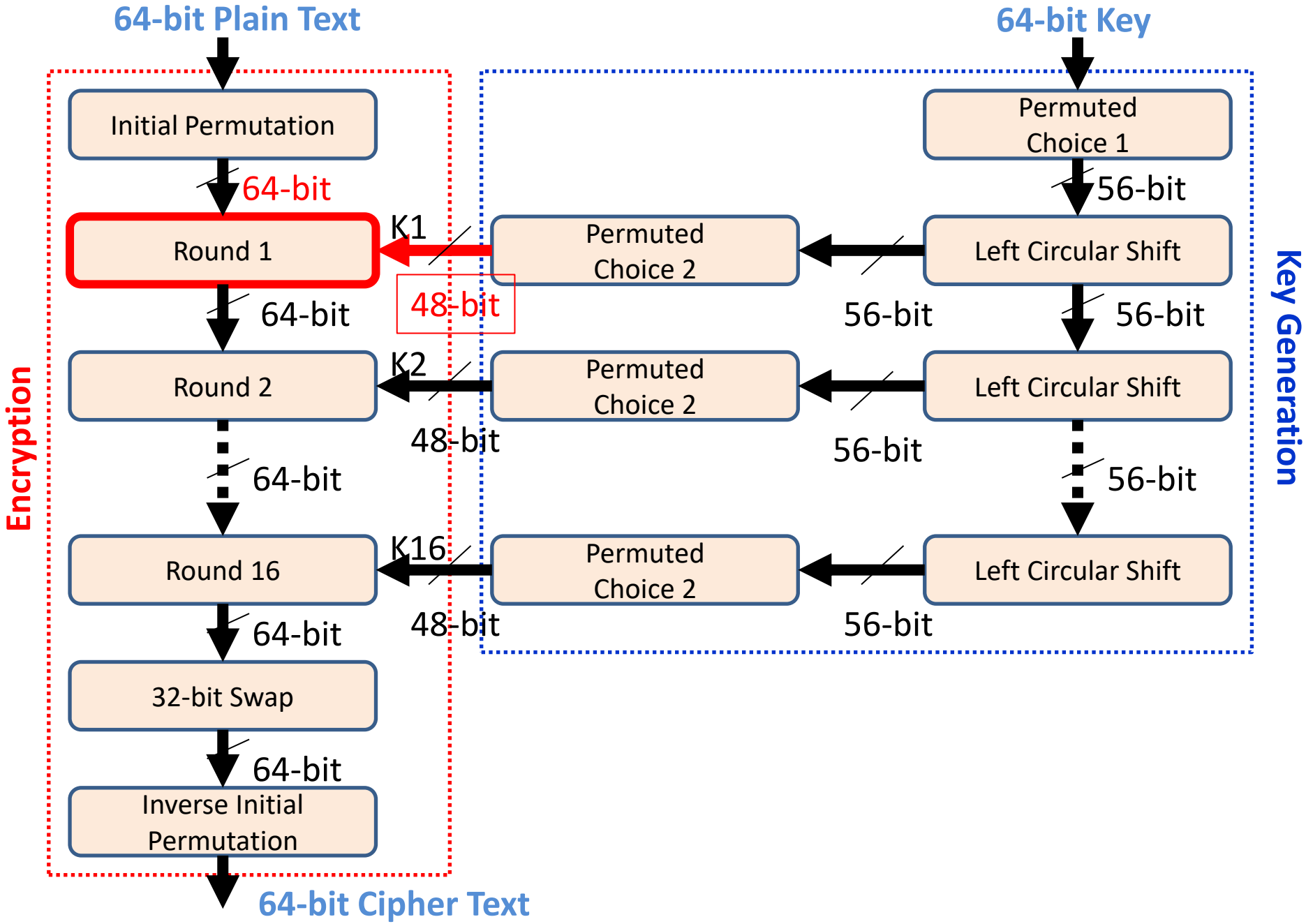
Initial Permutation table:

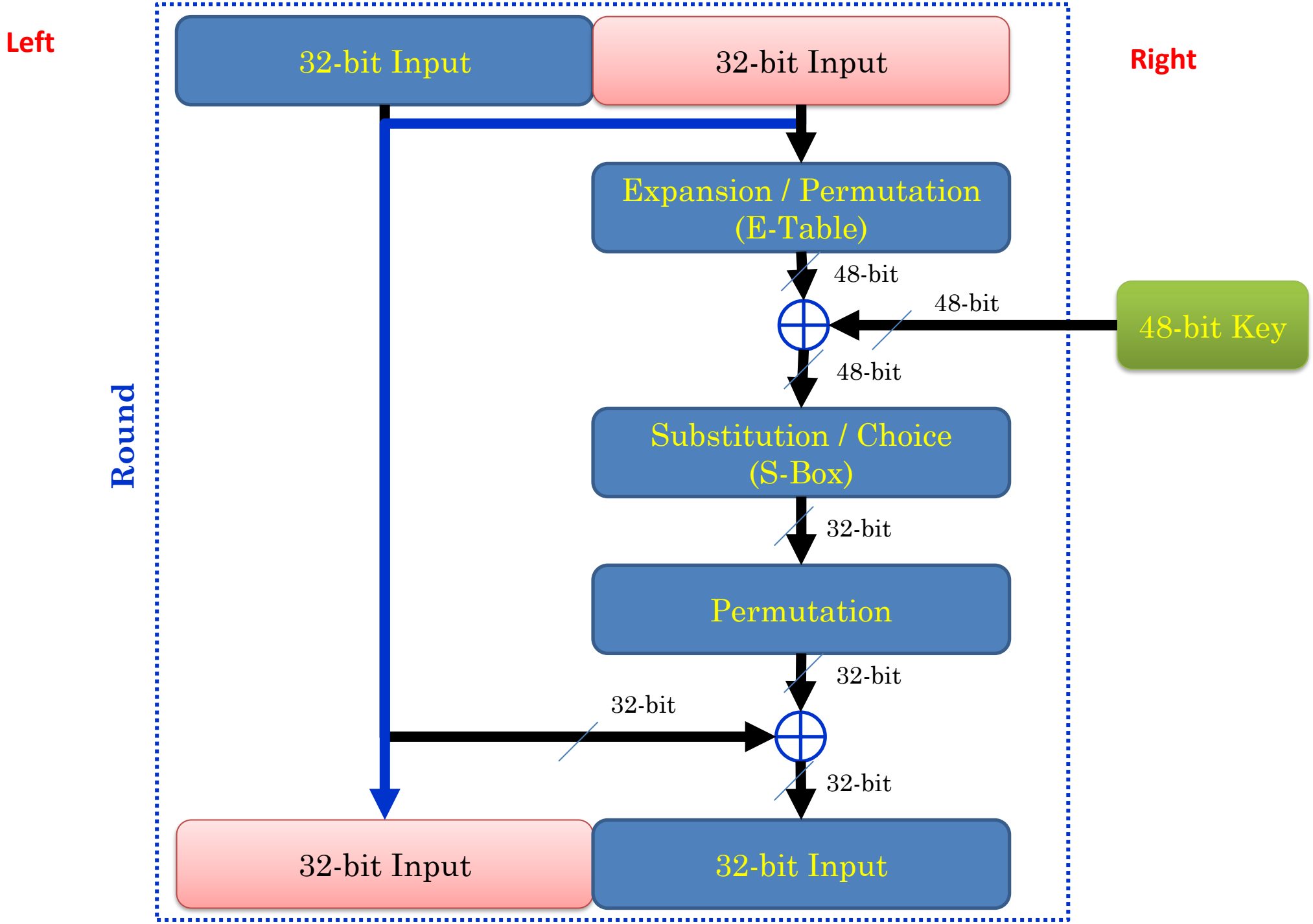
58	50	42	34	26	18	10	2
60	52	44	36	28	20	12	4
62	54	46	38	30	22	14	6
64	56	48	40	32	24	16	8
57	49	41	33	25	17	9	1
59	51	43	35	27	19	11	3
61	53	45	37	29	21	13	5
63	55	47	39	31	23	15	7

After Permutation :

1	1	1					

DES- Encryption




Left
Right
Round
32-bit Input
32-bit Input
Expansion / Permutation
(E-Table)
48-bit
48-bit
48-bit Key
48-bit
Substitution / Choice
(S-Box)
32-bit
Permutation
32-bit
32-bit
32-bit
32-bit Input
32-bit Input